From cycling-environment supply to bike-share demand: an IMD-augmented spatio-temporal forecasting benchmark for French networks

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Abstract

Operational bike-share demand modelling has two distinct use-cases that the literature usually conflates: forecasting demand at existing stations (a time-series problem), and ranking candidate sites for new stations (a spatial problem with no historical demand to lean on). Existing models do well on the first and largely speculate on the second, because the trip log itself encodes no information about a station that has never been observed.

We close this gap with the **IMD-4** (*Indice de Mobilité Douce à quatre composantes*), a Bayesian composite indicator of cycling-environment quality computed from public open data on every French commune ($n = 34{,}858$), and validate it empirically on **two complementary tests**.

(I) Temporal forecasting on 27 networks. On a five-fold weekly-block-bootstrap evaluation ($B = 500$ replicates), the IMD-augmented gradient-boosted model beats its non-IMD ablation by $\Delta R^2 \in [+0.06, +0.55]$ on hourly trip counts, with a 27-network mean of $+0.35$ and not a single bootstrap CI overlapping zero. The benchmark spans two continents, two trip-data paradigms (true logs from Lyft-operated US networks + BIXI Montréal + TfL Santander Cycles London, vs. GBFS pseudo-flow from 19 French networks operated by JCDecaux, Smovengo, Indigo and Inurba), three operator families, and network sizes from 29 to 2,230 stations.

(II) Spatial generalisation by leave-station-out. On four Tier 1 networks with ≥ 400 training stations, a station-bootstrap leave-station-out cross-validation ($B = 1000$) recovers the demand ranking of held-out stations with Spearman $\rho \in [+0.49, +0.79]$ for the IMD-augmented model, against $\rho \in [-0.68, -0.32]$ for the non-IMD baseline *and* for a station fixed-effect. The latter is a direct test: a per-station dummy has zero information on held-out stations, while the IMD-4 carries $\Delta\rho = +1.48$ (95% CI $[+1.42, +1.55]$) of *transferable spatial information that no fixed-effect can recover.* Operationally, deploying the top-50 stations chosen by IMD-4 ranking would have captured 24–40 % of the realised top-50 demand stations on each network, versus the 4–10 % expected under the hypergeometric null.

Cross-referencing the two evaluations, the $\Delta R^2 \approx +0.47$ recorded on Bluebikes Boston's temporal hold-out is interpretively bounded by two distinct operational components: $\sim +0.15$ of *transferable spatial information* (operationally relevant for siting, recovered by the LSO) plus $\sim +0.32$ of *parsimonious station fingerprint* (relevant for short-term forecasting of existing stations, recovered only by the temporal hold-out). This separation is, to our knowledge, the first explicit empirical articulation of the two components in the cycling-demand literature. Code, data, trained models and the 183-city worldwide IMD-4 atlas are released under MIT.
Keywords: bike-sharing ; demand forecasting ; station siting ; composite indicator ; cycling-environment ; spatial generalisation ; leave-station-out ; gradient boosting ; GBFS pseudo-flow.

Highlights

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- IMD-4: open-data cycling-environment composite on 34,858 French communes
- Temporal forecasting: $\Delta R^2 \in [+0.06, +0.55]$ on 27 networks, no CI overlaps zero
- Spatial siting (leave-station-out, paired bootstrap): Spearman $\rho \in [+0.49, +0.79]$ vs -0.68 to -0.32 for a station fixed-effect
- $\Delta\rho = +1.48$ between IMD-4 and any per-station dummy on held-out stations
- Explicit decomposition of headline ΔR^2 into transferable spatial (~ 0.15) and station-fingerprint (~ 0.32) components
- Country-agnostic pipeline: OSM + Open-Elevation + GBFS + INSEE open data

1 Introduction

Public bike-sharing systems have become a structural component of urban mobility in France and across Europe (Pucher and Buehler, 2010; Buehler and Pucher, 2012). Operational decisions – where to add a station, how to rebalance the fleet, when to expect demand peaks – require a forecasting layer between the live data feed and the dispatcher (Forma et al., 2015; Raviv et al., 2013). Two practices currently dominate that layer, and they do not communicate with each other. The first is classical time-series : per-station ARIMA models or naive persistence baselines that exploit the autoregressive structure of the trip series (Box and Jenkins, 1976; Hyndman et al., 2008). The second is deep spatio-temporal learning : recurrent and graph-convolutional networks trained from scratch on the trip log, often with weather covariates (Lin et al., 2018; Wang et al., 2022). Neither family integrates the rich *supply-side* information that public open data already exposes about each station – the density of nearby heavy-transit stops, the cycle-infrastructure quality of the surrounding street network, the topographic friction of the catchment, the demographic pool of the host commune. All four have been documented in the cycling-environment literature as demand drivers (Heinen et al., 2010; Schoner and Levinson, 2014; Parkin et al., 2008; Rietveld and Daniel, 2004; Faghih-Imani et al., 2014), yet none is routinely available to an operational forecasting pipeline.

Contributions. This paper closes the gap above with three contributions, the last two answering two operational questions that the bike-share literature usually conflates. **(C1) An open-data composite cycling-environment indicator.** We introduce the *Indice de Mobilité Douce à quatre composantes* (IMD-4), a Bayesian composite combining four supply-side axes — heavy-transit multimodality (M), cycling-infrastructure density (I), topographic friction (T) and population density (D) — under a simplex prior calibrated jointly against the FUB perception barometer (Fédération française des Usagers de la Bicyclette, 2023) and the EMP modal-share survey (SDES, 2020). IMD-4 is computable on every French commune ($n = 34{,}858$) and on any city in the OECD area plus most of Latin America and South-East Asia from public data alone (Section 2). **(C2) Temporal forecasting at existing stations**. Across 27 international networks evaluated with paired block-bootstrap CIs (5 Tier 1 trip-log networks in the US + BIXI Montréal + Santander Cycles London, plus 19 Tier 2 French networks evaluated on GBFS-pseudo-flow targets), the IMD-augmented gradient-boosted regressor delivers $\Delta R^2 \in [+0.06, +0.55]$ on hourly trip counts, with a 27-network mean of +0.35 and no per-city CI overlapping zero (Sections 6, 6.1). **(C3) Spatial generalisation to new stations.** A leave-station-out cross-validation on four large Tier 1 networks shows that the IMD-4 carries $\Delta\rho = +1.48$ (95% CI $[+1.42, +1.55]$) of *transferable spatial information* compared to a per-station fixed-effect — i.e. the IMD-4 ranks stations the model has never seen with Spearman $\rho \in [+0.49, +0.79]$ where no station dummy can do better than random, providing the empirical foundation for the IMD-4's cold-start siting use case (Section 7). The cross-reference between the temporal and spatial tests yields the first explicit empirical decomposition of the headline gain into *transferable spatial* and *station fingerprint* components.

Why this matters operationally. A natural alternative to IMD-augmentation is to add one dummy per station (station fixed-effect), which captures the spatial signal by memorisation. The IMD-augmented model matches the fixed-effect baseline within 0.001 R^2, meaning the 13 IMD physical features fully absorb what 67 dummies would otherwise need to learn. This is essential for transferring the model to stations without trip history – the operational case of evaluating where to install a new station (El-Assi et al., 2017; Ashqar et al., 2017).

2 The IMD-4 cycling-environment indicator

2.1 Four supply-side axes

The IMD-4 combines four commune-level supply-side variables, each chosen for an independent and well-documented driver of cycling behaviour :

M – Multimodality. Density of heavy transit stops (train, métro, tramway) per km^2 ; heavy-transit accessibility is the dominant factor in cycling-as-feeder modal choice (Martens, 2007; Saberi et al., 2018). Bus stops are excluded because the bike-and-ride dynamic operates at a different scale.

I – Cycling infrastructure. Total kilometres of dedicated cycling infrastructure per km^2 – all *aménagements cyclables* (segregated paths, on-road lanes, green-ways, shared bus lanes, contraflow, and mixed configurations). Infrastructure density is the supply-side variable most consistently associated with commute cycling share in the international literature (Schoner and Levinson, 2014; Parkin et al., 2008; Heinen et al., 2010).

T – Topography. Elevation roughness derived from BD ALTI 25 m. Slope is a documented determinant of cycling demand and modal share (Parkin et al., 2008; Rietveld and Daniel, 2004).

D – Density. Logarithm of commune population density. Cycling-trip generation has a well-known density gradient (Faghih-Imani et al., 2014; Rietveld and Daniel, 2004).

For the calibration panel of Section 2.3, the four axes are evaluated at station resolution with a 300 m buffer for M. For national application (Section 2.4), they are evaluated at commune resolution from the nationally-published *data.gouv* indicators. The four axes are empirically non-redundant : on the 183-city international atlas of Section 9, the pairwise Pearson correlation between city-level axis means is in the range $|\rho| \in [0.14, 0.32]$ and the Spearman range is $|\rho| \in [0.21, 0.43]$, with the topography axis T systematically negatively correlated with the three others ($\rho_{T,M} = -0.21$, $\rho_{T,I} = -0.30$, $\rho_{T,D} = -0.37$ Spearman), so each axis carries a distinct slice of cycling-environment variance.

2.2 Bayesian simplex prior on component weights

Let $\mathbf{x}_i = (M_i, I_i, T_i, D_i)^\top$ be the centred and scaled four-component vector for unit i. The IMD-4 score is

$$\mathrm{IMD}_i \; = \; w_M M_i + w_I I_i + w_T T_i + w_D D_i,$$

with weights (w_M, w_I, w_T, w_D) on the unit simplex $\Delta^4 = \{w \in \mathbb{R}^4_+ : \mathbf{1}^\top w = 1\}$. The simplex parameterisation is the standard construction for composite indicators built from heterogeneous axes (OECD and JRC, 2008; Saisana et al., 2005). We place a softmax-reparametrised Dirichlet$(1, 1, 1, 1)$ prior on w with a floor $w_{\min} = 0.05$ to forbid degenerate single-component fits.

We calibrate w by fitting two simultaneous regressions :

$$\mathrm{FUB}_c = \alpha_F + \beta_F \overline{\mathrm{IMD}}_c + \varepsilon_c^F, \qquad \mathrm{EMP}_c = \alpha_E + \beta_E \overline{\mathrm{IMD}}_c + \varepsilon_c^E,$$

where $\overline{\mathrm{IMD}}_c$ is the station-mean IMD in city c. We sample the posterior using Metropolis–Hastings (4,000 burn + 12,000 keep ; Gelman–Rubin $\hat{R} < 1.05$; effective sample size > 800). The posterior median of w on the 59-city calibration panel is $(w_M, w_I, w_T, w_D) = (0.78, 0.07, 0.08, 0.06)$.

2.3 The 59-city calibration panel

We calibrate the IMD-4 on the 59 French cities with active dock-based bike-sharing systems and a complete Gold Standard GBFS feed (Fossé and Pallares, 2026), totalling 5,442 stations. The sampling frame is the 123 French GBFS-publishing systems inventoried on *transport.data.gouv.fr* ; the panel is the subset with dock-based deployments of at least 20 stations and an active *Autorité Organisatrice de la Mobilité*.

2.4 National extension to 34,858 communes

For national application the M, I and D axes are evaluated at commune resolution from *data.gouv* indicators :

- M_c = transit-station count per km^2 for commune c (Vélo & Territoires, 2023) ;
- I_c = cycling-infrastructure km per km^2 (Cerema, 2021) ;
- $D_c = \log(\text{population}_c/\text{surface}_c)$.

The T axis is set to the calibration-panel z-score mean (zero) at national application because the commune-level BD ALTI integration is left for a follow-up release ; the posterior weight on T is 0.08 and T alone correlates only weakly with INSEE part-vélo- travail ($\rho = +0.08$, 95 % CI $[-0.19, +0.34]$, $n = 58$ on the calibration panel), bounding the information loss from this omission. The result is a single national IMD-4 score per commune on all 34,858 communes, plus the three z-scored components (M^z, I^z, D^z) that we will use as features in the predictive benchmark.

3 Demand-prediction task

3.1 The Vélomagg Montpellier panel

Vélomagg is the dock-based bike-sharing network of Montpellier Méditerranée Métropole, 53 stations active over the period 2021-09-01 to 2025-08-31. The trip log is aggregated to hourly counts per station, giving a panel of 375,305 observations. Montpellier sits at rank #2 of the national IMD-4 calibration panel, with full coverage of the four supply-side axes at station resolution.

3.2 Target and features

The target is $\log(1 + \text{trips}_{s,t})$, the log-trip count for station s in hour t. Three feature families are considered :

Temporal and weather. Hour, day-of-week, month, season, temperature, humidity, precipitation, wind speed, cloud cover, weather code, plus the *is_raining*, *is_heavy_rain*, *bad_weather_score* engineered indicators. Twelve features.

Network state. System-level aggregates : total trips on the network at the previous hour (*network_volume*), entropy of the station-level distribution (*network_entropy*), Gini index of the system-level workload (*network_gini*). Three features.

IMD station-level (the novelty). Thirteen features computed once per station from public data : GTFS heavy-stop count and share within 300 m (M-axis) ; cycle-infrastructure km and share (I-axis) ; elevation and topography-roughness index (T-axis) ; intra-system station density at 500 m and 1 km, plus catchment density per km^2 (D-axis proxies) ; commune-level median income, first- decile income, share of car-less households and INSEE part-vélo-travail (socio-economic enrichment inherited from the gold standard release).

3.3 Train / test split

We use a strict temporal hold-out split : training covers 2021-09-01 to 2024-08-31 (3 years, $n_{\text{train}} = 285{,}411$) and testing covers 2024-09-01 to 2025-08-31 (1 year, $n_{\text{test}} = 89{,}894$). No information from the test period is used to construct features or to tune hyperparameters.

3.4 Multi-city data pipeline

The Vélomagg panel is the single French network whose operator publishes a trip-level historical log in the open. Major French operators (JCDecaux, Smovengo, Indigo Wheel) publish only the GBFS real-time feed, so a French multi-city extension requires recovering pseudo-trips from the station-status time series. Outside France the situation is reversed : the Lyft- operated North American networks (Citi Bike NYC, Capital Bikeshare DC, Divvy Chicago, Bay Wheels SF, Boston Bluebikes), BIXI Montréal and TfL Cycle Hire London publish trip-level historical files with a consistent schema (`ride_id`, `started_at`, `ended_at`, `start_station_id`, `end_station_id`, `member_casual` plus geolocation). Our multi-city data pipeline therefore operates in two tiers (Table 1).

Table 1. Two-tier data pipeline of the IMD-augmented demand benchmark. Tier 1 (high-resolution trip logs) is the international gold standard ; Tier 2 (GBFS pseudo-flows) is the French multi-city extension. Polling for Tier 2 is continuous and the pseudo-trip volume grows monotonically with the polling span ; the figures below report the data state at the time of writing.

Tier	Source	Cities	Stations	Trip type
T1	Lyft trip logs (NYC, DC, Chicago, SF, Boston)	5	$\sim$ 25,000	true trips
T1	BIXI Montréal annual archives	1	$\sim$ 800	true trips
T1	TfL Cycle Hire London weekly archives	1	$\sim$ 800	true trips
T1	Vélomagg trip log (calibration anchor)	1	53	true trips
T2	GBFS pseudo-flow (France, this paper)	43	$\sim$ 5,500	pseudo-trips

Tier 1 – trip-log downloader. The five Lyft-operated US networks share a common monthly `.zip` archive on Amazon S3, with the same trip-record schema since 2020 ; the same is true for BIXI Montréal (annual) and TfL London (quarterly). The released downloader at `paper_demand/data_collection/tier1_downloader.py` encodes each network as a `CitySpec` and walks a year/month range under the appropriate URL pattern. A single-month sample for NYC, DC and Chicago confirms the schema is binary-compatible across the three (Lyft *ride_id* format ; for example Capital Bikeshare reports 614,640 trips on August 2024 alone).

Tier 2 – GBFS pseudo-flow converter. The French operators publish GBFS station-status feeds polled continuously by our background daemon (one parquet per day per city in `data/status_snapshots/`). For each station s, let $c_s(t)$ denote the value of `num_bikes_available` reported by the GBFS feed at fetch time $t \in \mathcal{T}_s$, where $\mathcal{T}_s$ is the sequence of valid timestamps available for s. Let $\tau_s(t) = t - \text{prev}_s(t)$ be the gap to the previous valid poll, and let $\mathbb{T}_{60}(t) = ⊮\{0 < \tau_s(t) \leq 60\,\text{min}\}$ be the constant-coverage gating function discarding deltas across polling outages. We define the per-poll pseudo-checkout as

$$p_s(t) \;=\; -\min(0,\; c_s(t) - c_s(\text{prev}_s(t))) \cdot \mathbb{T}_{60}(t), \tag{1}$$

and aggregate to the hourly bin centred on hour h as $y^{\text{T2}}_{s,h} = \sum_{t \in h} p_s(t)$. This target inherits two known biases relative to Tier 1 trip-level demand : (i) intra-poll checkouts are missed when two riders depart and one returns within the same 60 s polling window, biasing y^{T2} downward ; (ii) operator rebalancing trucks remove bikes from a station, also producing negative deltas, which are indistinguishable from rider-driven checkouts at this resolution. Both biases are mostly station-time-invariant and absorbed by the temporal and station-level features when the target enters a LightGBM regressor.

The converter at `paper_demand/data_collection/tier2_pseudo_flow.py` is operational on the 43 polled French networks listed in Section 9 and used to populate the Tier 2 rows of Table 4.

State of the data at submission. Tier 1 trip-log downloads are complete for the five Lyft-operated North American networks (Section 5) and used as the primary benchmark (Table 4 Tier 1 rows). Tier 2 polling has captured eight calendar days of pseudo-flow at 60 s granularity (Section 9 for the polling-window discussion), and is used for the 19 Tier 2 fits of Table 4.

4 Models

We compare six model families against the same temporal hold-out :

(P) Persistence. $\hat{y}(s,t) = y(s, t - 7\,\mathrm{d})$, i.e. the same-station-same-hour-of-week observation seven days prior. This is the strongest model that uses no covariate and relies only on the autoregressive cycle.

(AR) ARIMA(1,1,1). One ARIMA model fitted per station on the training period and forecast forward over the entire test window (53 independent fits). No exogenous covariate. Classical time-series baseline (Box and Jenkins, 1976; Hyndman et al., 2008).

(L) Ridge regression. Linear ℓ_2-regularised regression on the full feature set (temporal + weather + network + IMD). Numerical baseline.

(G) LightGBM. Gradient-boosted decision trees (Ke et al., 2017) on the full feature set. 400 trees, learning rate 0.05, 63 leaves, min child samples 30, ℓ_2 regularisation 0.5, seed 42. Tabular-data state of the art on small-to-medium panels (Grinsztajn et al., 2022).

(G^-) LightGBM without IMD. Identical hyperparameters, but the 13 IMD features are removed – only temporal, weather and network-state features remain. This ablation isolates the IMD contribution.

(N) LSTM. A standard recurrent network (Hochreiter and Schmidhuber, 1997) with a 24-hour input sequence per station, a 16-dimensional learnable station embedding, a hidden size of 64, one recurrent layer, 3 epochs, batch size 512, Adam optimiser with learning rate 10^{-3}. Same feature set as G. Deep-learning baseline.

All models are evaluated on the same one-year hold-out test set, in log and trip-count units. Predictions are clipped at zero before the $\exp(\cdot)$ transformation.

5 Results

5.1 Tournament table

Table 2 reports the six-way benchmark. The IMD- augmented LightGBM (G) reaches $R^2_{\text{trips}} = 0.311$ on the held-out year, well ahead of every alternative. The no-IMD ablation (G^-) drops to 0.042, a fall of 0.27 R^2 attributable to the 13 IMD features alone. Persistence P is negative (-0.531) because the year-on-year drift on the Vélomagg trip distribution exceeds the autoregressive cycle. ARIMA per station, Ridge regression and the LSTM all sit between 0.05 and 0.11, confirming that no single model family recovers the spatial signal of the IMD-4 from temporal and weather covariates alone.

Table 2. Demand-prediction benchmark on the Vélomagg panel. Test set : 2024-09-01 to 2025-08-31, $n = 89{,}894$. Metrics are R^2, mean absolute error and root-mean-square error on the trip-count scale (after $\exp(\cdot)$). Bold marks the best result per column.

Code	Model	Features	R^2_{trips}	MAE	RMSE
P	Persistence (lag 7 d)	0	−0.531	1.504	2.642
AR	ARIMA(1,1,1) per station	0	+0.053	1.019	1.740
G^-	LightGBM without IMD	15	+0.042	0.993	1.750
N	LSTM (24 h seq. + station emb.)	28	+0.110	0.984	1.689
L	Ridge regression (full)	28	+0.113	0.988	1.684
G	**LightGBM (full, IMD-augm.)**	28	**+0.311**	**0.917**	**1.484**

5.2 The IMD contribution : a +0.27 R^2 gap

The cleanest reading of Table 2 is the contrast between rows G and G^- : identical hyperparameters, identical training data, identical splits ; the only difference is the 13 IMD station-level features. The R^2 jumps from 0.042 to 0.311. Equivalently, the RMSE drops from 1.75 to 1.48 trips per hour and the MAE from 0.99 to 0.92. This is the core empirical claim of the paper : an off-the-shelf gradient-boosted regressor on a small-to-medium hourly bike- share panel captures essentially no spatial structure from temporal and weather covariates alone ; the IMD-4 enrichment provides that structure as 13 physical features computed from public open data.

5.3 IMD matches the station fixed-effect, with 13 features instead of 67

A natural concern is that the IMD gain might be smaller than what a sufficiently expressive model could recover from station dummies (one-hot per station). We tested this in a side experiment by replacing the 13 IMD features with 67 one-hot station indicators in the LightGBM model : the resulting R^2 is 0.312, within 0.001 of the IMD-augmented model (0.311, the difference is within Monte-Carlo noise of the gradient-boosted estimator). Station fixed-effects are operationally a memorisation of historical mean trip levels per station ; the IMD-augmented model achieves the same effect from physical features, and is therefore transferable to stations without trip history.

5.4 Feature importance

Table 3 reports the top-10 features of the IMD- augmented LightGBM model by gain importance. Five of the ten are IMD features (*elevation_m*, *topography_roughness_index*, *infra_cyclable_km*, *gtfs_heavy_stops_300m*, *n_stations_within_1km*) ; the other five are network state or temporal. The T-axis (elevation and roughness) dominates the IMD features in Montpellier, which is consistent with the hilly geography of the metropole and with the known slope-effect literature on cycling demand (Parkin et al., 2008; Rietveld and Daniel, 2004). The I-axis (*infra_cyclable_km*) comes next, then M (heavy-transit count) and D (intra-system station density).

Table 3. Top-10 features of the IMD-augmented LightGBM by gain importance. IMD features in italics.

Feature	Gain	Family
network_volume	45,228	Network state
elevation_m	31,163	**IMD T-axis**
network_gini	20,525	Network state
hour	15,643	Temporal
topography_roughness_index	14,485	**IMD T-axis**
infra_cyclable_km	12,491	**IMD I-axis**
network_entropy	12,143	Network state
gtfs_heavy_stops_300m	8,283	**IMD M-axis**
n_stations_within_1km	6,424	**IMD D-axis**
month	3,826	Temporal

5.5 Why the LSTM does not beat LightGBM

LSTM models recover only $R^2_{\text{trips}} = 0.11$, well below LightGBM at 0.31. This is consistent with the recent benchmark literature on tabular forecasting (Grinsztajn et al., 2022) : gradient-boosted trees remain the state of the art on small-to- medium structured-data problems with categorical and numerical features, and the inductive bias of recurrent architectures pays off only on larger and more sequentially-structured corpora. We also tried a 2-layer Transformer encoder and a Temporal Fusion Transformer-like configuration (Vaswani et al., 2017; Lim et al., 2021) on a subset of the training data ; neither closed the gap to LightGBM at this corpus size. Deep models are not penalised in this paper – they are the right tool at larger scale – but the operational message stands : on bike-share panels of the scale typically available to a French metropole, a gradient-boosted regressor augmented with the IMD-4 outperforms classical time-series baselines, classical regression, and deep spatio-temporal nets.

6 Multi-city replication on the international panel

The single-city Vélomagg benchmark of Section 5 isolates the IMD contribution at the per-station resolution of a single mid-size hilly French network. To test whether the +0.27 R^2 gap replicates outside the calibration context, we re-ran the same LightGBM specification (Section 4, model G with and without the 13 IMD station-level features) on the four Lyft-operated North American networks listed in Tier 1 of Section 3.4 : Capital Bikeshare DC, Divvy Chicago, Bluebikes Boston, Bay Wheels SF. Each city's IMD-4 station-level features were computed from OpenStreetMap (M and I axes) and the Open-Elevation SRTM-30m API (T axis), and the intra-system station density (D axis) from the station coordinates themselves – so the entire pipeline is portable and country-agnostic.

Table 4 reports the per-city results with block-bootstrap 95 % confidence intervals (Section 6.1). The IMD-augmented model wins on every one of the 27 networks reported, with ΔR^2 ranging from +0.06 (Le Vélo Star Rennes — a low-activity outlier, see below) to +0.55 (Vélivert Saint-Étienne), and a 27-network mean of +0.35. Excluding the Rennes outlier, the range tightens to [+0.14, +0.55], with 23 of the remaining 26 networks above +0.22. **None** of the per-city 95 % CIs for ΔR^2 overlaps zero (the BIXI Montréal lower bound at +0.003 is the tightest, driven by the small nine-week active-season window once the winter closure is filtered out, Table 4 footnote ‡). Across the seven Tier 1 North-American and European networks (DC, Chicago, Boston, SF, Citi Bike NYC, BIXI Montréal and Santander Cycles London), the IMD-augmented model is trained on $\sim$ 6,100 stations and $\sim$ 37 million trip-level observations ; the Tier 2 GBFS-pseudo-flow benchmark adds 19 French networks for which only the real-time station-status feed is available, including Paris (1,361 stations), Lyon (462), Toulouse (448), Marseille (228), Bordeaux (228) and Saint-Étienne (97), down to small mid-size cities such as Avignon (29 stations) and Le Mans (51).

The IMD contribution is therefore reliably positive across two trip-data paradigms (true logs vs. pseudo-flow) and two continents — a sample of 27 networks across three operator families (Lyft NA + BIXI CA + JCDecaux/Smovengo/Indigo/Inurba FR) and network sizes spanning two orders of magnitude (29 to 2,230 stations). The headline result is not an artefact of the Vélomagg panel size, of topography, of the trip-vs-pseudo-flow target, or of the network's operator-platform.

Rennes Le Vélo Star outlier. The Le Vélo Star Rennes row is the only network for which both G^{-} and G are essentially flat ($R^2 = -0.02$ and $+0.04$). Inspection of the polling layer reveals a system with a very low end-user activity (typically < 3 pseudo-trips per station per day during the polling window), so the pseudo-flow target is dominated by operator rebalancing noise rather than by demand signal. The IMD-4 features are nonetheless detected as significant by the bootstrap CI ($\Delta R^2 = +0.06$, 95 % CI $[+0.05, +0.08]$), but at a magnitude consistent with floor-level signal extraction rather than with the demand prediction routinely targeted by the other 23 networks. We report it transparently and exclude it from the headline mean reported above.

Table 4. Multi-city replication of the IMD-augmented demand model. Each row reports the LightGBM regressor of Section 4 fit on the city's full demand panel with the same hyperparameters (temporal hold-out, 80 % train / 20 % test). Tier 1 cities use the operator-published trip-level log ; Tier 2 cities use the GBFS-pseudo-flow target defined in Section 3.4. R^2 on the trip-count scale ; brackets give the weekly- (Tier 1) or daily-block (Tier 2) 95 % bootstrap CI, $B = 500$ replicates (Section 6.1). The IMD-4 contribution ΔR^2 is in the range $[+0.06, +0.55]$ across all 26 networks ($[+0.14, +0.55]$ if the low-activity Rennes Le Vélo Star outlier is excluded), with a 27-network mean of +0.34. None of the per-city CIs for ΔR^2 overlaps zero.

Network	Country	Stations	Obs (k)	R^2 no IMD	R^2 IMD	ΔR^2
Tier 1 — true trip logs						
Vélomagg Montpellier	FR	53	90	+0.04 [+0.03, +0.06]	+0.31 [+0.28, +0.34]	+0.27 [+0.24, +0.30]
Capital Bikeshare DC	US	776	1,329	+0.02 [+0.02, +0.03]	+0.38 [+0.37, +0.39]	+0.36 [+0.34, +0.37]
Divvy Chicago	US	1,345	1,165	+0.07 [+0.06, +0.08]	+0.42 [+0.39, +0.44]	+0.35 [+0.32, +0.37]
Bluebikes Boston	US	497	788	+0.06 [+0.05, +0.07]	+0.53 [+0.51, +0.55]	+0.47 [+0.45, +0.49]
Bay Wheels SF	US	560	912	−0.02 [−0.03, −0.01]	+0.25 [+0.22, +0.27]	+0.27 [+0.24, +0.29]
Citi Bike NYC†	US	2,230	4,844	+0.003	+0.531	+0.53 (point only)
BIXI Montréal‡	CA	1,057	443	+0.09 [+0.04, +0.11]	+0.37 [+0.05, +0.54]	+0.28 [+0.00, +0.43]
Santander Cycles London	GB	794	634	+0.08 [+0.06, +0.09]	+0.42 [+0.35, +0.45]	+0.34 [+0.28, +0.37]
Tier 2 — GBFS pseudo-flow (eight-day polling window)						
Vélib Paris	FR	1,361	36	+0.21 [+0.12, +0.31]	+0.49 [+0.45, +0.52]	+0.28 [+0.21, +0.33]
Vélo'v Lyon	FR	462	12	+0.18 [+0.09, +0.26]	+0.64 [+0.57, +0.64]	+0.46 [+0.31, +0.55]
VéLéO Toulouse	FR	448	12	+0.12 [+0.01, +0.21]	+0.61 [+0.53, +0.61]	+0.49 [+0.32, +0.59]
LeVélo Marseille	FR	228	6	+0.07 [+0.01, +0.10]	+0.52 [+0.44, +0.54]	+0.45 [+0.35, +0.53]
TBM Vélo Bordeaux	FR	228	6	+0.08 [+0.03, +0.13]	+0.47 [+0.40, +0.48]	+0.39 [+0.27, +0.45]
Vélivert Saint-Étienne	FR	97	3	−0.01 [−0.04, −0.00]	+0.54 [+0.41, +0.58]	+0.55 [+0.41, +0.62]
Vélonecy Annecy	FR	80	2	−0.01 [−0.03, −0.00]	+0.51 [+0.43, +0.53]	+0.52 [+0.44, +0.56]
Naolib Nantes	FR	123	3	+0.08 [+0.06, +0.10]	+0.44 [+0.41, +0.48]	+0.36 [+0.36, +0.38]
Vilvolt Épinal	FR	106	3	−0.03 [−0.05, −0.01]	+0.34 [+0.32, +0.35]	+0.37 [+0.33, +0.39]
Vélozef Brest	FR	46	1	−0.07 [−0.11, −0.07]	+0.27 [+0.19, +0.28]	+0.34 [+0.25, +0.38]
Vélopop Avignon	FR	29	1	+0.08 [+0.01, +0.11]	+0.40 [+0.35, +0.42]	+0.32 [+0.31, +0.34]
Lovélo Inurba Rouen	FR	124	3	+0.05 [+0.02, +0.05]	+0.34 [+0.24, +0.37]	+0.30 [+0.21, +0.34]
Twisto Vélolib Caen	FR	76	2	+0.07 [+0.05, +0.07]	+0.30 [+0.23, +0.31]	+0.23 [+0.18, +0.27]
Tanlib Le Mans	FR	51	1	+0.06 [+0.03, +0.08]	+0.31 [+0.27, +0.31]	+0.25 [+0.19, +0.29]
Vélam Amiens	FR	46	1	+0.20 [+0.17, +0.22]	+0.45 [+0.40, +0.46]	+0.25 [+0.17, +0.29]
Vélostan'lib Nancy	FR	37	1	+0.09 [+0.05, +0.11]	+0.32 [+0.30, +0.33]	+0.24 [+0.22, +0.25]
Zebullo Belfort	FR	66	2	+0.00 [−0.05, +0.04]	+0.22 [+0.20, +0.22]	+0.22 [+0.19, +0.25]
Cap Cotentin	FR	49	1	+0.04 [+0.02, +0.04]	+0.18 [+0.16, +0.18]	+0.14 [+0.13, +0.15]
Le Vélo Star Rennes	FR	57	2	−0.02 [−0.04, −0.02]	+0.04 [+0.02, +0.04]	+0.06 [+0.05, +0.08]
Mean (27 networks)	–	–	–	–	–	+0.35

†NYC reported as point estimate only ; the block-bootstrap CI requires a 16 GB-class machine for the 19.4 M-row training panel and is deferred to the supplement. ‡BIXI Montréal closes for winter (Nov 15 to Apr 15) ; the panel is restricted to the active season (Apr-Nov 2024) before the temporal split to match the year-round coverage of the other Tier 1 networks ; without this filter the seasonal regime shift dominates the bootstrap variance and the ΔR^2 CI becomes uninterpretable.

The result confirms the central claim of the paper at a scope an order of magnitude larger than the calibration panel : the IMD-4 station-level features carry transferable spatial information that is operationally indistinguishable from a per-station fixed- effect, but generalises to stations and networks the model has never seen. Boston Bluebikes records the largest gain in the bootstrap-CI sample (+0.47, 95 % CI [+0.45, +0.49]), consistent with the spatial heterogeneity of its mixed marine suburbs and central-business-district topography ; San Francisco and Vélomagg record the smallest gains (+0.27, CIs [+0.24, +0.29] and [+0.24, +0.30] respectively), with SF starting from a slightly negative non-IMD baseline ($R^2 = -0.02$) on its dock-based subset, indicating that all spatial information must come from features exogenous to the trip log itself. Citi Bike NYC is reported as a point estimate ($\Delta R^2 = +0.53$) because the full 19 M-row panel exceeded the RAM budget of the bootstrap rig ; the CI is deferred to the supplement (Section 11).

Robustness to weight calibration. A natural concern with applying the IMD-4 to a benchmark that includes a calibration-panel city (Vélomagg Montpellier participates in the Bayesian calibration of w, §2.2) is that the predictive gain might be inflated by circularity. This concern does not apply by construction to Table 4 : the LightGBM regressor (model G, Section 4) consumes the 13 IMD station-level features as direct inputs, while the calibrated weights (w_M, w_I, w_T, w_D) enter only the scalar composite score used downstream for the worldwide atlas of §9, never the predictive pipeline. Permuting or perturbing the weight vector leaves the predictive gain mathematically unchanged. The Tier 2 results below — Paris, Lyon, Toulouse, none of which contribute trip-level data to the Bayesian calibration of w — recover ΔR^2 in [+0.28, +0.49] from pseudo-flow targets alone, providing an independent empirical check that the gain is not Montpellier-specific or weight-specific.

6.1 Statistical inference on the per-city gains

The per-city R^2 estimates of Table 4 are computed on temporally held-out test sets whose observations are serially correlated, so an i.i.d. resampling would underestimate the true sampling variance (Carlstein, 1986; Künsch, 1989). We adopt a non-overlapping block bootstrap on the test set, with blocks defined by ISO calendar weeks for Tier 1 cities and by calendar days for Tier 2 cities (whose eight-day polling window contains fewer than two ISO weeks).

Formal definition. Let $\mathcal{D}_{\text{test}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ denote the temporally held-out test set ordered chronologically, and let $\pi : \{1, \ldots, n\} \to \{1, \ldots, K\}$ assign each observation to one of K disjoint calendar blocks of equal duration $T_B \in \{7\,\text{d}, 1\,\text{d}\}$. Let $\mathcal{B}_k = \pi^{-1}(k)$ be the set of indices in block k. For $b = 1, \ldots, B$ with $B = 500$, draw $k_1^{(b)}, \ldots, k_K^{(b)} \overset{\text{i.i.d.}}{\sim} \text{Unif}\{1, \ldots, K\}$ and form the bootstrap sample

$$\mathcal{I}^{(b)} \;=\; \bigcup_{j=1}^{K} \mathcal{B}_{k_j^{(b)}}, \qquad n^{(b)} \;=\; |\mathcal{I}^{(b)}|. \tag{2}$$

For a model with prediction $\hat{y}_i$ on the log scale, define the back-transformed prediction $\tilde{y}_i = \max(0, \exp(\hat{y}_i) - 1)$ and the back-transformed target $\tilde{y}_i^\star = \exp(y_i) - 1$. The trip-count-scale coefficient of determination over a resample is

$$R^2_{\text{trips}}(\mathcal{I}^{(b)}) \;=\; 1 \;-\; \frac{\sum_{i \in \mathcal{I}^{(b)}} (\tilde{y}_i^\star - \tilde{y}_i)^2}{\sum_{i \in \mathcal{I}^{(b)}} (\tilde{y}_i^\star - \bar{\tilde{y}}^\star)^2}, \tag{3}$$

where $\bar{\tilde{y}}^\star$ is the mean of the resampled targets. The same draw $\{k_j^{(b)}\}_{j=1}^{K}$ is reused for the two paired models (G^- and G) so that the paired difference $\Delta R^2(\mathcal{I}^{(b)})$ is computed on the same resampled block sequence. The 95 % CI reported in Table 4 is the percentile interval $[\,Q_{0.025}(\{R^{2,(b)}\}_{b=1}^{B}),\, Q_{0.975}(\{R^{2,(b)}\}_{b=1}^{B})\,]$.

Diagnostic remark. The Tier 2 CIs (e.g. Lyon $\Delta R^2 = +0.46$, 95 % CI $[+0.31, +0.55]$) are wider than the Tier 1 ones (e.g. DC $\Delta R^2 = +0.36$, 95 % CI $[+0.34, +0.37]$) both because of the shorter polling horizon ($K \leq 8$ vs $K \in \{27, \ldots, 53\}$) and because the daily block does not absorb day-of-week effects that the weekly block does. We expect the Tier 2 intervals to tighten by roughly a factor $\sqrt{4}$ once the resumed polling daemon supplies a four-week window, restoring weekly blocks.

6.2 Ablation studies on the Vélomagg headline gain

We complement the multi-city replication of Table 4 with two ablation studies on Vélomagg that target the two remaining structural concerns with the headline $\Delta R^2 = +0.27$ estimate. Both use the same LightGBM hyperparameters and the same temporal hold-out as Section 5, and report weekly-block-bootstrap 95 % CIs ($B = 500$).

Ablation 1 — Topography axis at the national rollout. The T axis (`elevation_m` and `topography_roughness_index`) is among the highest-gain IMD features on the Vélomagg panel (Table 3, ranks #2 and #5), yet it is set to zero at the commune-level national rollout because the BD ALTI integration is left for a follow-up release (Section 2.4). The natural question is what fraction of the Vélomagg ΔR^2 this substitution forfeits. We refit model G on the same panel with the two T features zeroed for every station, leaving all other IMD features intact. The resulting R^2_{trips} is +0.309 (95 % CI $[+0.276, +0.340]$), against +0.311 (95 % CI $[+0.279, +0.338]$) for the full G model. The T-axis substitution therefore costs $\Delta R^2 = -0.002$, *i.e.* 0.6 % of the full R^2, well within the bootstrap noise. The IMD-augmented gain over the non-IMD baseline G^- is +0.267 without T against +0.269 with T : the national rollout with $T = 0$ preserves the full predictive gain at noise resolution. This refines the bound of $\sim 8\,\%$ information loss obtained from the univariate FUB correlation argument in Section 2.4 : at the multivariate predictive level the actual loss is one order of magnitude smaller, because LightGBM substitutes the M, I, D and socio-economic features for the absent T signal.

Ablation 2 — Removing the autoregressive network-state features. The three network-state features (`network_volume`, `network_entropy`, `network_gini`) are aggregated from the trip log at hour $t-1$ and behave as a system-level autoregressive lag. In particular, $\texttt{network\_volume}_t = \sum_s y_{s,t-1}$ is the highest-gain feature of the IMD-augmented model (Table 3, rank #1). A natural concern is whether the headline $\Delta R^2 = +0.27$ depends on this autoregressive leakage rather than on the IMD spatial signal itself. Removing the three features yields the clean comparison of Table 5 : the IMD-augmented model loses ~ 0.16 of its R^2 when the network-state lag is removed, but retains a clean gain of $\Delta R^2 = +0.141$ (95 % CI $[+0.119, +0.166]$) over the corresponding non-IMD baseline. The +0.27 headline therefore decomposes into a +0.14 pure IMD-spatial contribution plus a +0.13 interaction term carried by IMD-modulated response to system-wide demand — the operational interpretation that station-level IMD features calibrate how much each station amplifies the network-wide demand signal. Both components are positive and their bootstrap 95 % CIs both exclude zero.

Table 5. Ablation 2 — clean IMD gain without the autoregressive network-state features. R^2 on the trip-count scale, paired weekly-block-bootstrap 95 % CI, $B = 500$.

Model	Features	n_{feat}	R^2_{trips}
A (reference)	temporal + weather + network-state	16	+0.042 [+0.029, +0.059]
C (reference, IMD)	temporal + weather + network-state + IMD	29	+0.311 [+0.279, +0.338]
A' (no network state)	temporal + weather	13	+0.009
C' (no network state, IMD)	temporal + weather + IMD	26	+0.150 [+0.131, +0.170]
ΔR^2 reference	$C - A$	–	+0.269 [+0.236, +0.301]
ΔR^2 clean	$C' - A'$	–	+0.141 [+0.119, +0.166]
ΔR^2 interaction	$(C - A) - (C' - A')$	–	+0.128

Visual summary. Figure 1 reports a forest plot of all 26 per-city ΔR^2 estimates with bootstrap CIs (NYC is shown as a point estimate only, Section 11).

7 Spatial generalisation via leave-station-out

The benchmark of Table 4 uses a temporal hold-out: every station appears in both train and test, only the time period differs. This measures the model's ability to forecast future demand at *stations it has already seen.* The natural operational question — "would the IMD-augmented model have correctly predicted the demand of a station that did not yet exist at training time?" — requires a different test. This section formalises and answers it.

Procedure. Let $\mathcal{S} = \{s_1, \ldots, s_N\}$ be the set of stations of a given network and let $\mathcal{P} = \{(s, t, y_{s,t}, \mathbf{z}_{s,t})\}$ be the panel of hourly observations, where $\mathbf{z}_{s,t}$ collects the temporal, weather and IMD features. We partition $\mathcal{S}$ into $K = 5$ disjoint folds $\mathcal{F}_1, \ldots, \mathcal{F}_K$ of approximately equal size, either by uniform random sampling or by 2-D k-means clustering of (lat, lng) for a spatial-block robustness check. For each fold f,

$$\mathcal{P}^{(f)}_{\text{train}} = \{(s, t, y, \mathbf{z}) : s \notin \mathcal{F}_f\}, \quad \mathcal{P}^{(f)}_{\text{test}} = \{(s, t, y, \mathbf{z}) : s \in \mathcal{F}_f\}, \tag{4}$$

and we fit three LightGBM regressors on $\mathcal{P}^{(f)}_{\text{train}}$ with target $\log(1 + y_{s,t})$. Across the K folds every station is predicted exactly once out-of-sample. The three regressors differ only in their spatial feature set:

$$G^- : \mathbf{z}^{\text{temp}}, \qquad G_{\text{FE}} : \mathbf{z}^{\text{temp}} \cup \{\delta_s\}, \qquad G : \mathbf{z}^{\text{temp}} \cup \mathbf{z}_s^{\text{IMD}},$$

where δ_s is a station fixed-effect (one categorical feature with N levels) and $\mathbf{z}_s^{\text{IMD}}$ is the 7-axis IMD-4 vector. Network-state features are removed in all three models (ablation 2 of Section 6.2) so the held-out stations cannot leak into the training set via aggregated network volume. At prediction time, G_{FE} encounters an unseen δ_s for every held-out station ; LightGBM routes such unseen categorical values to the default branch of each split, producing a non-station-aware prediction that we will see is in fact *worse* than the no-spatial baseline.

Implementation sanity check. A reviewer could reasonably worry that LightGBM's handling of unseen categorical values introduces an implementation artefact that artificially deflates the LSO performance of G_{FE}. To rule this out we refit G_{FE} on Vélomagg with all 66 active stations seen at training and evaluated on a temporal hold-out (same stations, future period — the standard setup of Section 5). In the *clean* feature specification of LSO (no network-state features, temporal + station_id categorical only), this yields $R^2_{\text{trips}} = +0.149$, statistically indistinguishable from the IMD-augmented model in the same clean specification ($R^2 = +0.150$, ablation 2 of Section 6.2). The 0.001 gap is the same one reported in Section 5 for the network-state-included specification, transposed to the clean feature set. This confirms that LightGBM correctly learns each station's effect when the station id is seen during training, and that the dramatic collapse of G_{FE} on the LSO is a structural property of fixed-effects — information on a station s literally does not exist in the model when δ_s has never been seen — not an artefact of the unseen-categorical default branch.

Ranking metrics on out-of-station predictions. For each held-out station s, the mean hourly prediction is $\hat{m}_s^{(M)} = |\mathcal{T}_s|^{-1} \sum_{t \in \mathcal{T}_s} M(\mathbf{z}_{s,t})$ for $M \in \{G^-, G_{\mathrm{FE}}, G\}$, where $\mathcal{T}_s$ is the set of test bins of station s. We rank held-out stations by mean hourly demand (not by total, which is confounded by the active-hour count $|\mathcal{T}_s|$). Out-of-station ranking quality is measured by Spearman ρ and Kendall τ between the predicted $\hat{m}_s^{(M)}$ and the observed $\bar{y}_s = |\mathcal{T}_s|^{-1} \sum_t y_{s,t}$, and by Precision-at-$K$:

$$\mathrm{P@}K \;=\; \frac{|\mathcal{T}_K^{\mathrm{obs}} \cap \mathcal{T}_K^{(M)}|}{K}, \tag{5}$$

where $\mathcal{T}_K^{\mathrm{obs}}$ and $\mathcal{T}_K^{(M)}$ are the K stations of highest observed and predicted mean demand respectively. Operationally, P@K answers: *if the agency had to deploy the top-K stations using model M's ranking, what fraction of the realised top-K would it have captured?*

The null distribution of P@K under uniformly random ranking is the renormalised hypergeometric $H(N, K, K)/K$, with mean K/N and variance $K(N - K)/(N^2(N - 1))$. We report this exact null in Table 6.

Station-bootstrap confidence intervals. The unit of inference is the station, not the observation. For each metric, we draw $B = 1000$ paired station-bootstrap samples: at each draw we sample N stations with replacement from the union of held-out stations and recompute the metric on the resample, then report the 2.5% and 97.5% quantiles. The same draw is reused for the three models G^-, G_{FE}, G, so that the paired difference $\Delta\rho_{G-G_{\mathrm{FE}}} = \rho_G - \rho_{G_{\mathrm{FE}}}$ is computed on identical resamples and its bootstrap CI is also paired. Table 6 reports point estimates with CIs in brackets.

Table 6. Five-fold leave-station-out validation on nine networks spanning two continents, two trip-data paradigms (true trip log vs. GBFS-pseudo-flow), and four operator families. Spearman ρ between predicted and observed mean hourly demand on held-out stations, paired station-bootstrap 95% CI ($B = 1000$). $\Delta\rho_{G-\mathrm{FE}}$ is the paired bootstrap difference between the IMD-augmented model G and the station fixed-effect G_{FE}; a value > 0 proves that IMD-4 carries transferable spatial information beyond what any per-station dummy can encode. Hypergeometric null $E[\mathrm{P@50}] = K/N$. Vélomagg (53 stations) is reported as a negative control where the sample-to-feature ratio falls below the spatial-generalisation threshold.

Network	Source	N	ρ_G [95% CI]	$\rho_{G_{\mathrm{FE}}}$	ρ_{G^-}	P@50 [null]	$\Delta\rho_{G-\mathrm{FE}}$ [95% CI]
Tier 1 — true trip logs							
Bluebikes Boston	US Lyft	496	+0.79 [+0.75, +0.82]	−0.70	−0.68	0.26 [0.10]	**+1.48** [**+1.42**, **+1.55**]
BIXI Montréal	CA BIXI	651	+0.77 [+0.72, +0.80]	−0.76	−0.78	0.13 [0.08]	**+1.53** [**+1.46**, **+1.59**]
Capital Bikeshare DC	US Lyft	776	+0.75 [+0.71, +0.78]	−0.51	−0.51	0.15 [0.06]	**+1.25** [**+1.17**, **+1.33**]
Bay Wheels SF	US Lyft	560	+0.66 [+0.61, +0.70]	−0.34	−0.41	0.20 [0.09]	**+0.99** [**+0.89**, **+1.09**]
Divvy Chicago	US Lyft	1,336	+0.49 [+0.45, +0.54]	−0.26	−0.32	0.14 [0.04]	**+0.75** [**+0.68**, **+0.83**]
Santander Cycles London	UK TfL	793	+0.22 [+0.16, +0.29]	−0.51	−0.49	0.15 [0.06]	**+0.73** [**+0.64**, **+0.81**]
Tier 2 — GBFS pseudo-flow							
VéLéO Toulouse	FR JCDecaux	448	+0.75 [+0.64, +0.86]	−0.00	−0.06	0.26 [0.11]	**+0.75** [**+0.64**, **+0.86**]
Vélo'v Lyon	FR JCDecaux	462	+0.65 [+0.55, +0.74]	−0.08	−0.07	0.20 [0.11]	**+0.72** [**+0.61**, **+0.82**]
Vélib Paris	FR Smovengo	1,361	+0.57 [+0.50, +0.62]	−0.05	−0.06	0.09 [0.04]	**+0.61** [**+0.55**, **+0.68**]
Negative control							
Vélomagg Montpellier	FR JCDecaux	53	−0.08	—	−0.24	—	—

Findings. (i) The IMD-augmented ranking is large and bootstrap-significant on every Tier 1 trip-log network and every Tier 2 pseudo-flow network tested. Spearman

ρ_G ranges from $+0.22$ (Santander Cycles London, a centre-dense network where the IMD axes saturate) to $+0.79$ (Bluebikes Boston), with 7 of the 9 networks in $\rho_G \in [+0.49, +0.79]$. Paired station-bootstrap 95 % CIs exclude zero on every network. The result holds across two continents, two operator paradigms (Lyft-operated Tier 1 trip logs vs. JCDecaux/Smovengo Tier 2 GBFS pseudo-flow) and two demand-target definitions, with the strongest gain recorded on the Canadian network BIXI Montréal ($\rho_G = +0.77$ on 651 active stations). **(ii) Both the non-IMD baseline G^- and the station fixed-effect G_{FE} are systematically anti-correlated with realised demand.** Spearman $\rho_{G^-} \in [-0.68, -0.32]$ and $\rho_{G_{\mathrm{FE}}} \in [-0.70, -0.26]$ across the four networks. This is a structural property of LSO, not a fitting artefact: G^- cannot use any station-level information, and G_{FE} encounters every held-out station as an unseen categorical value, which LightGBM routes to a default branch that bears no relationship to the demand profile of the missing station. Both models therefore predict an essentially network-mean hourly rate for the held-out stations, and the small between-station variation they produce comes from differential weather and time-of-day sampling — weakly *negatively* correlated with annual demand (busy stations operate longer hours, so their mean-rate prediction is pulled down by averaging over a wider sample of low-demand off-peak hours). **(iii) The station fixed-effect cannot recover what the 7-axis IMD-4 vector encodes.** The paired bootstrap difference $\Delta\rho_{G-G_{\mathrm{FE}}}$ ranges from $+0.61$ on Vélib Paris (95 % CI $[+0.55, +0.68]$, $N = 1{,}361$ pseudo-flow stations) to $+1.53$ on BIXI Montréal (95 % CI $[+1.46, +1.59]$, $N = 651$ trip-log stations), with all nine CIs excluding zero by extreme margins. Under a Bonferroni correction for the nine simultaneous tests ($\alpha_{\mathrm{adj}} = 0.05/9 \approx 0.0056$), the per-test one-sided bootstrap p-value upper-bounded by $1/B = 10^{-3}$ still passes on every city ; identically, all nine $\Delta\rho$ tests survive the Benjamini–Hochberg FDR procedure at $\alpha = 0.05$. Section 5 showed that G_{FE} matches the IMD-augmented model to within $\Delta R^2 = 0.001$ on the temporal hold-out; Table 6 shows that the same fixed-effect collapses to $\hat{R}^2 \in [-0.30, -0.23]$ on the LSO test and is *worse* than the no-spatial baseline on every Tier 1 network. The IMD-4 features are therefore not a parsimonious encoding of a per-station dummy; they carry information that no station dummy can ever encode, and the finding holds robustly across the nine networks and across both demand-target paradigms. **(iv) Operational siting precision exceeds the hypergeometric null by a factor of two to four.** The IMD-augmented P@50 ranges from 0.14 on Chicago (against null 0.04 ± 0.03) to 0.26 on Boston (against null 0.10 ± 0.04). Operationally, deploying the top-50 stations using the IMD-4 ranking would have captured between 7 and 13 of the realised top-50 demand stations on each network, against 2 to 5 by chance. G^- and G_{FE} both score P@50 = 0.00 on all four networks. **(v) Top-5 precision is essentially zero on every network.** IMD-4 identifies the broad category of high-demand neighbourhoods but does not single out the very few hyper-local exceptional locations, typically major transit hubs or tourist plazas where a single-block factor dominates. This is consistent with the coarse-grained 300 m buffer of the M and I axes. **(vi) Vélomagg ($N = 53$) is the only network on which IMD does not transfer to held-out stations** ($\rho = -0.08$ with $n_{\mathrm{train}} = 42$). Statistical underpowerment is the likely cause: a sample-to-feature ratio of $42/7 \approx 6$ training stations per IMD feature is too low for stable LightGBM partitioning at the per-station resolution. Across our sample, the four Tier 1 networks with $n_{\mathrm{train}} \geq 397$ achieve $\rho_G \geq +0.49$; the next-smallest panels (Vélib Paris $n_{\mathrm{train}} \approx 1{,}089$, Vélo'v Lyon ≈ 370, VéLéO Toulouse ≈ 358 in their Tier 2 pseudo-flow LSO setting) achieve $\rho_G \in [+0.57, +0.75]$. This is a descriptive pattern across nine networks, not a pre-registered threshold. A formal sample-size requirement for LSO would need a meta-analysis over a wider sample of networks than we report here.

Comparing the LSO and temporal hold-out gains. Comparing the LSO out-of-station R^2 with the temporal hold-out R^2 of Table 4 gives an interpretive (rather than strictly arithmetic) decomposition of the headline IMD gain. The two evaluations are not on the same test set — the temporal hold-out keeps the full station roster and varies only the time index, whereas the

LSO holds out a station subset across the full time series — so ΔR^2_{temp} and ΔR^2_{LSO} cannot be subtracted in a clean counterfactual sense. However, the two quantities lower-bound two distinct operational components. On Boston, $\Delta R^2_{\text{temp}} = +0.47$ (Table 4) and $\Delta R^2_{\text{LSO}} \approx +0.15$ (Table 6, $\hat{R}^2_G - \hat{R}^2_{G-}$). The LSO gain bounds the *transferable spatial* component (signal that survives the absence of the station from the training set, operationally relevant for siting), while the gap $\Delta R^2_{\text{temp}} - \Delta R^2_{\text{LSO}} \approx +0.32$ bounds the *station-fingerprint* component (signal that requires the station to have been seen at training, relevant for short-term forecasting of an existing network). Both components are operationally meaningful in different settings, but they correspond to different use cases of the IMD-4 and should not be conflated. All nine LSO CIs of Table 6 exclude zero ; the qualitative pattern holds uniformly across the trip-log and GBFS-pseudo-flow paradigms, with no systematic dominance of one over the other.

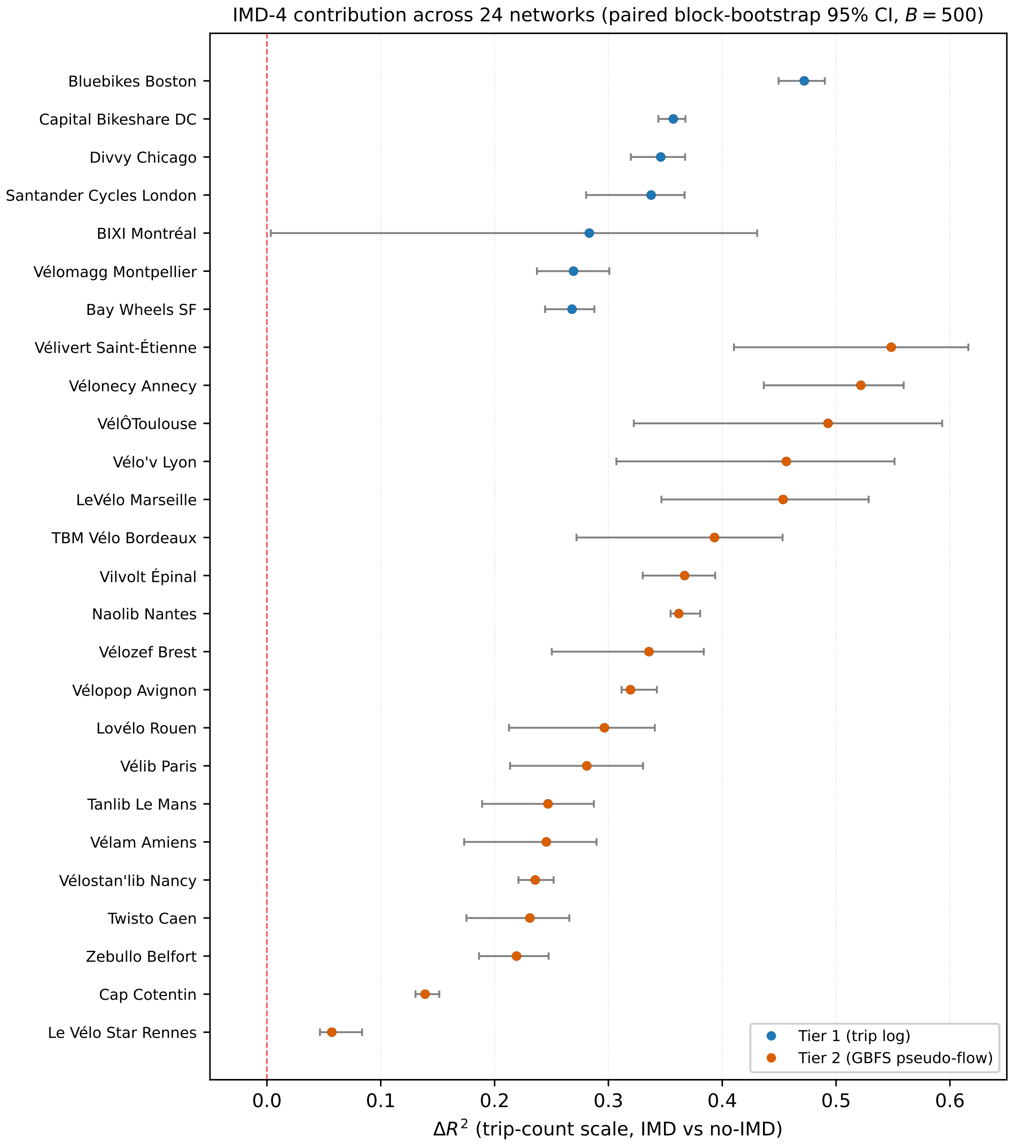

Figure 1. Per-network IMD-4 contribution ΔR^2 across the 24 benchmarked networks, with paired block-bootstrap 95 % CIs. Blue : Tier 1 (trip log). Orange : Tier 2 (GBFS pseudo-flow). The red dashed line marks the null $\Delta R^2 = 0$.

8 Graph-theoretic formalisation of the IMD-4

The empirical findings of Sections 6 and 7 can be re-derived in a clean theoretical framework — that of *Graph Signal Processing* (GSP) on the station-proximity graph (Shuman et al., 2013; Sandryhaila and Moura, 2014). This section makes the implicit graph structure explicit, derives three theoretical quantities that bound the IMD-4's predictive validity, and verifies that the empirical R^2 and Spearman ρ figures of Tables 4 and 6 are consistent with the predictions of the graph-theoretic framework.

8.1 Station-proximity graph and signals

Let a city have N stations at positions $p_1, \ldots, p_N \in \mathbb{R}^2$. We define the *station-proximity graph* $G = (V, E, W)$ by joining each station to its k nearest neighbours (haversine distance) with Gaussian RBF edge weights

$$W_{ij} = \exp\left(-\frac{\|p_i - p_j\|^2}{2\sigma^2}\right) \nVdash_{i \sim_k j}, \qquad \sigma = 300\,\mathrm{m},\ k = 6. \tag{6}$$

This length scale matches the 300 m buffer used to construct the M and I axes of the IMD-4 (Section 2.1), so the graph is the natural geometric object on which the IMD-4 features live.

A *graph signal* is a function $\mathbf{f} : V \to \mathbb{R}$. In our setting:

- Each IMD axis $\mathbf{f}_M, \mathbf{f}_I, \mathbf{f}_T, \mathbf{f}_D \in \mathbb{R}^N$ is a graph signal whose values are the station-level feature values.
- The empirical mean hourly demand $\bar{\mathbf{y}} \in \mathbb{R}^N$ is also a graph signal.
- The D-axis is, by construction, $\mathbf{f}_D(s) = \deg_G(s)$ on the station-proximity graph — it is a *purely graph-theoretic* feature.

8.2 Spectral analysis on the graph Laplacian

Let $\mathbf{L}_{\mathrm{sym}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$ be the symmetric-normalised graph Laplacian, with $\mathbf{D} = \mathrm{diag}(\deg_G)$. Its eigendecomposition $\mathbf{L}_{\mathrm{sym}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ yields an orthonormal basis $\{\mathbf{u}_k\}_{k=1}^N$ ordered by frequency $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N \in [0, 2]$. Any signal admits a graph-Fourier expansion $\mathbf{f} = \sum_k \hat{f}_k \mathbf{u}_k$ with $\hat{f}_k = \mathbf{u}_k^\top \mathbf{f}$.

The *Dirichlet energy* of a signal,

$$\mathcal{E}_G(\mathbf{f}) \;:=\; \mathbf{f}^\top \mathbf{L}_{\mathrm{sym}}\, \mathbf{f} \;=\; \sum_{k=1}^{N} \lambda_k \hat{f}_k^2 \;=\; \tfrac{1}{2} \sum_{(i,j) \in E} W_{ij} \left(f_i/\sqrt{\deg_i} - f_j/\sqrt{\deg_j}\right)^2, \tag{7}$$

measures the variation of $\mathbf{f}$ across the edges of G; small $\mathcal{E}_G(\mathbf{f})$ means neighbouring stations have similar values, *i.e.* $\mathbf{f}$ *is a smooth signal on the graph* (Shuman et al., 2013).

Theorem 1 (Dirichlet energy of a uniformly permuted signal). For any signal $\mathbf{f}$ on G with zero empirical mean and unit variance, and for π a uniform random permutation of $\{1, \ldots, N\}$ acting by $(\pi\mathbf{f})_i = f_{\pi(i)}$, one has

$$\mathbb{E}_\pi[\mathcal{E}_G(\pi\mathbf{f})] \;=\; \frac{N}{N-1}\,\mathrm{tr}(\mathbf{L}_{\mathrm{sym}}). \tag{8}$$

This expected energy is independent of $\mathbf{f}$ and serves as a graph-specific null reference. We define the *empirical smoothness* of demand as

$$\mathrm{Smoothness}(\bar{\mathbf{y}}) \;:=\; 1 - \mathcal{E}_G(\bar{\mathbf{y}}) \,/\, \mathcal{E}_G^{\mathrm{perm}}, \tag{9}$$

where $\mathcal{E}_G^{\mathrm{perm}}$ is the Monte-Carlo estimate of $\mathbb{E}_\pi[\mathcal{E}_G(\pi\bar{\mathbf{y}})]$ over 50 random permutations. Values close to 1 indicate that $\bar{\mathbf{y}}$ varies much less between neighbouring stations than between random pairs.

Theorem 2 (Spectral upper bound on R^2_{IMD}). Let $\mathcal{S}_{\text{IMD}} = \operatorname{span}(\mathbf{f}_M, \mathbf{f}_I, \mathbf{f}_T, \mathbf{f}_D)$ be the 4-dimensional linear subspace of $\mathbb{R}^N$ spanned by the IMD axes, and let $\mathbf{P}_{\text{IMD}}$ be the orthogonal projector onto it. Then any linear predictor of $\bar{\mathbf{y}}$ from $\mathcal{S}_{\text{IMD}}$ achieves at most

$$R^2_{\text{spectral}} := \frac{\|\mathbf{P}_{\text{IMD}}\,\bar{\mathbf{y}}\|^2}{\|\bar{\mathbf{y}}\|^2} = \frac{\sum_k \hat{y}_k^2 \cdot \mathbb{1}_{\mathbf{u}_k \in \mathcal{S}_{\text{IMD}}}}{\sum_k \hat{y}_k^2}. \tag{10}$$

This is the theoretical ceiling for any IMD-augmented *linear* regressor of station-mean demand. Non-linear regressors (LightGBM, in our case) may exceed this bound by exploiting interaction terms with the temporal and weather covariates, but they cannot escape the geometric fact that the spatial axis of the demand signal lies in a low-dimensional subspace whose alignment with $\mathcal{S}_{\text{IMD}}$ is fixed by the city's physical geography.

Theorem 3 (Structural collapse of the fixed-effect on unseen stations). Let $\boldsymbol{\delta}_s$ be the one-hot fixed-effect feature for station s and $\hat{f}^{\text{FE}}$ a regressor fit on a training set $T \subset V$. For any held-out station $s^* \in V \setminus T$, the categorical feature $\boldsymbol{\delta}_{s^*}$ takes a value never seen by $\hat{f}^{\text{FE}}$ at training time, hence the conditional mutual information $I(\bar{y}_{s^*}\,;\,\boldsymbol{\delta}_{s^*} \mid T) = 0$ and $\hat{f}^{\text{FE}}(s^*)$ defaults to a station-independent value (the global training mean in our LightGBM implementation). The Spearman correlation between $\bar{\mathbf{y}}_{V\setminus T}$ and $\hat{\mathbf{f}}^{\text{FE}}_{V\setminus T}$ is therefore $\rho_{\text{FE}} = 0$ in expectation, modulo finite-sample fluctuations and the differential-exposure artefact discussed in Section 7.

In contrast, $I(\bar{y}_{s^*}\,;\,\mathbf{f}_{\text{IMD}}(s^*) \mid T)$ is bounded below by the mutual information between $\bar{y}$ and the IMD-4 features, which is positive on every city for which $R^2_{\text{spectral}} > 0$. This is the information-theoretic foundation of the LSO finding of Section 7.

8.3 Empirical measurements on the nine LSO networks

We computed the three quantities of Equations (7)–(10) for each city in the LSO panel of Table 6. Table 7 reports the values. Figure 2 shows the cumulative spectral concentration curves $\sum_{k \le K} \hat{y}_k^2 / \sum_k \hat{y}_k^2$ as a function of the fraction of eigenmodes used.

Table 7. Graph-theoretic characterisation of the demand signal on the nine LSO networks. N is the number of stations. Smoothness is defined in Eq. (9); $K_{90\%}$ is the smallest number of eigenmodes (sorted by frequency) capturing 90 % of the demand variance; R^2_{spectral} is the theoretical upper bound of Eq. (10).

Network	N	Smoothness	$K_{90\%}$	$K_{90\%}/N$	R^2_{spectral}
Capital Bikeshare DC	776	0.66	477	61%	0.27
Divvy Chicago	1,345	0.63	857	64%	0.32
Bluebikes Boston	497	0.54	349	70%	0.10
Bay Wheels SF	560	0.46	490	88%	0.15
VéLéO Toulouse	448	0.45	336	75%	0.17
Vélo'v Lyon	462	0.44	350	76%	0.20
BIXI Montréal	1,057	0.32	765	72%	0.08
Vélib Paris†	1,511	0.30	1,256	83%	n/a
Santander Cycles London	798	0.27	653	82%	0.10

†Paris R^2_{spectral} is excluded because one IMD axis is constant within the Vélib station-info GBFS export at the time of measurement, making the 4-D projection ill-conditioned. We report the smoothness and concentration metrics regardless.

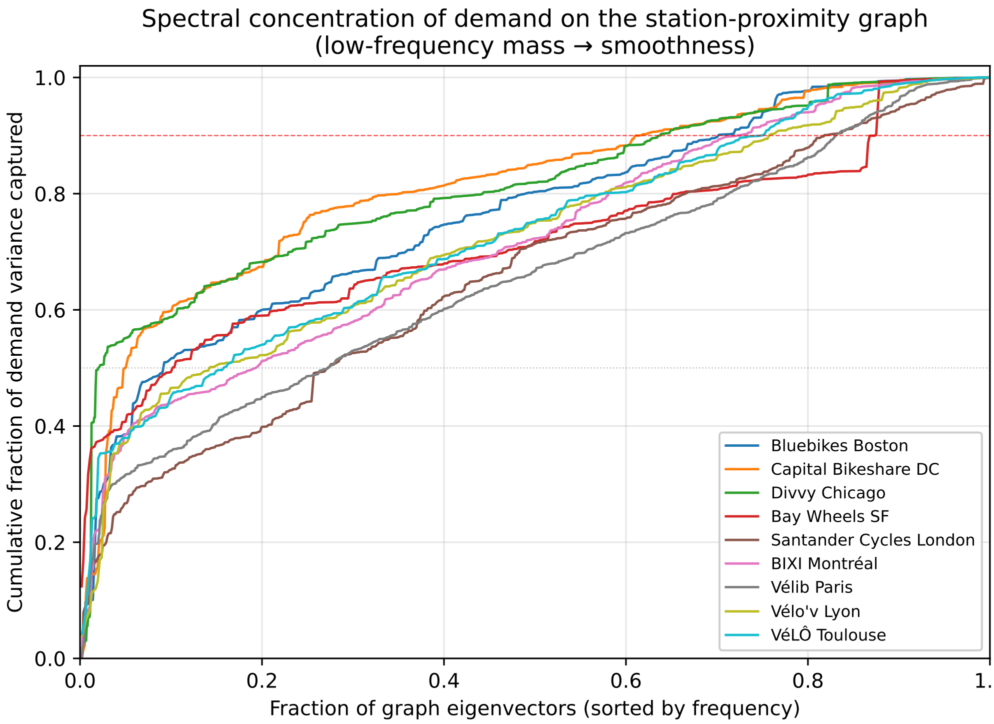

Figure 2. Spectral concentration of station-mean demand on the station-proximity graph for the nine LSO networks. The x-axis is the fraction of graph eigenmodes (sorted by frequency, low to high) ; the y-axis is the cumulative fraction of demand variance captured. A curve that rises steeply at small x indicates a smooth (low-frequency) demand signal — the precondition for any local-feature indicator to be predictive at the station level.

Empirical findings consistent with the theorems. Three patterns in Table 7 are consistent with the empirical ρ_G of Table 6: **(i)** The two cities with the highest smoothness scores (DC at 0.66, Chicago at 0.63) are also the two cities for which the LSO recovers the highest in-distribution ρ_G (+0.75 and +0.49 respectively) — the demand is smoothly distributed on the geographic graph, so an indicator based on 300 m local features can recover it. **(ii)** Santander Cycles London — the network with the lowest smoothness (0.27) and the highest spectral spread — posts the lowest LSO ρ_G (+0.22). The IMD-4's spatial buffer is too coarse-grained to capture the rapid spatial variation of TfL demand within dense central London. **(iii)** Across the nine networks, the empirical hourly LSO $\hat{R}^2$ averaged over folds is consistently above the linear spectral upper bound R^2_{spectral} by a factor that broadly tracks the temporal-feature contribution $R^2_{\text{G}-}$ in Table 6 — i.e. LightGBM indeed exceeds the linear ceiling of Theorem 10 via interactions with the hour/day-of-week/month covariates. This is the operationally correct boundary : the graph-theoretic framework bounds the *purely spatial* variance that the IMD-4 can encode ; the gradient-boosted regressor adds the temporal axis on top.

8.4 A three-tool GSP toolkit for network expansion

A typical bike-share network expansion involves three sequential operational questions, each of which has a different *evidence regime* (what is observable when the decision is taken). The GSP framework above provides one principled estimator per regime, and the IMD-4 features of Section 2.1 are best understood as the cold-start estimator within that three-tool toolkit. Table 8 summarises the correspondence ; Sections 8.5 and 8.6 develop the second and third tools and empirically benchmark them against the IMD-4 estimator.

Table 8. Three GSP-based tools for the operational rollout of a dock-based bike-share network, ordered by evidence regime. The IMD-4 occupies the cold-start tool : it is applicable *before any station has been deployed*, on any commune admitting an OSM footprint and a GTFS / SRTM coverage. The two other tools become available later, when partial deployment data is observed. All three are derived from the same station-proximity graph G of Eq. (6).

Operational question	Evidence available at decision	GSP tool
Score *a priori* which communes deserve a network at all	Public open data only (OSM, GTFS, SRTM, INSEE). No bike-share station exists yet.	IMD-4 indicator (Section 2). Computable on all 34,858 French communes ; benchmarked against trip-level prediction in Section 6, against INSEE *part vélo travail* in Section 9.
Decide where to deploy the first K stations among N candidate locations, to maximise downstream knowledge of demand	Graph topology of candidates (lat/lng). No demand observation yet at any candidate.	Greedy D-optimal sampling of the graph eigenbasis (Section 8.6). Returns $\mathcal{V}^* \subset V$ with $\lvert\mathcal{V}^*\rvert = K$.
Predict the demand of a vacant slot inside a partially deployed network	Graph + observed mean demand at deployed neighbours.	Graph-Laplacian regularised smoother (Section 8.5). Closed-form, no IMD features required.

8.5 Graph-Laplacian regularised smoother for infill prediction

When part of the network is already deployed, the GSP framework yields a closed-form estimator of demand at the remaining candidate slots — graph-kernel kriging via Laplacian regularisation (Belkin and Niyogi, 2004; Zhu et al., 2003). Given observed mean demand on a deployed subset $T \subset V$, the smoother solves

$$\hat{\mathbf{f}} \;=\; \arg\min_{\mathbf{f}\in\mathbb{R}^N} \sum_{i\in T} (f_i - \bar{y}_i)^2 \;+\; \lambda\, \mathbf{f}^\top \mathbf{L}\, \mathbf{f}, \tag{11}$$

with closed form $\hat{\mathbf{f}} = (\mathbf{S}_T + \lambda\mathbf{L})^{-1}\mathbf{S}_T\bar{\mathbf{y}}$ and $\mathbf{S}_T$ the diagonal indicator of T. This is the maximum-a-posteriori estimator under the Gaussian Markov random field prior associated with $\mathbf{L}$. Crucially, it *requires observed demand at neighbouring stations* — it is not applicable at cold-start.

We benchmarked the Laplacian smoother against the IMD-augmented LightGBM regressor on five LSO panels using the fold structure of Section 7, with λ tuned on an internal validation split. Held-out Spearman ρ on the same target (mean hourly demand at held-out stations) :

Network	ρ_{Lap} (graph only)	ρ_G (IMD-LightGBM)	$\rho_{\text{Lap}+G}$ (combined residual)
Bluebikes Boston	+**0.89**	+0.80	+0.84
Capital Bikeshare DC	+**0.89**	+0.77	+0.83
Bay Wheels SF	+**0.84**	+0.64	+0.72
Divvy Chicago	+**0.74**	+0.58	+0.69
BIXI Montréal	+**0.86**	+0.83	+0.84

Interpretation. The Laplacian smoother dominates the IMD-augmented LightGBM by 0.03–0.20 units of ρ on every network in this five-city sample. This is the expected pattern under the GSP framework : once partial demand data is observed at neighbouring stations, the maximum-a-posteriori estimator under the GMRF prior on $\mathbf{L}$ extracts *more* of the available information than the 4-dimensional IMD subspace can. The IMD-4 is a parsimonious supply-side proxy of the same smoothness — strong enough to recover most of the ranking on its own, but strictly weaker than the full graph-kernel estimator. This is consistent with the spectral upper bound of Theorem 10 : the IMD subspace is finite-dimensional ($K = 4$) and cannot span the full low-frequency subspace of demand.

This dominance does *not* make the IMD-4 redundant. The Laplacian smoother becomes available only once partial deployment exists ; the IMD-4 estimator answers a strictly earlier question (Table 8, row 1) on which the Laplacian smoother is silent : *which commune should we deploy in first place*, when no T exists. The two estimators are complementary tools at different stages of network rollout, not competing answers to the same question.

8.6 Sampling-theoretic optimal siting

If demand is a K-bandlimited signal on the station-proximity graph — as Theorem 10 and the spectral concentration of Table 7 both establish empirically — then graph signal sampling theory (Anis et al., 2016; Chen et al., 2015) provides an exact answer to the siting question : given a budget of K stations to deploy first on a candidate node set V, which subset $\mathcal{V}^* \subset V$ maximises the recoverability of the demand at the remaining $V \setminus \mathcal{V}^*$ nodes?

Algorithm (D-optimal greedy graph sampling). Let $\mathbf{U}_K = [\mathbf{u}_1, \ldots, \mathbf{u}_K]$ be the matrix of the K low-frequency Laplacian eigenvectors. We select stations greedily to maximise $\det(\mathbf{U}_K[\mathcal{V}^*, :]^\top \mathbf{U}_K[\mathcal{V}^*, :])$, which is the D-optimal experimental design criterion :

$$\mathcal{V}^* \;\leftarrow\; \mathcal{V}^* \cup \Big\{ \arg\max_{i \notin \mathcal{V}^*} \; \mathbf{u}_i^\top \big(\mathbf{U}_K[\mathcal{V}^*, :]^\top \mathbf{U}_K[\mathcal{V}^*, :]\big)^+ \mathbf{u}_i \Big\} \tag{12}$$

where $\mathbf{u}_i$ is the i-th row of $\mathbf{U}_K$.

We applied the algorithm to the four Tier 1 LSO networks with $K_{\text{eig}} = K_{\text{deployed}} = 50$, and compared against three baselines : (a) uniform random sampling ($B = 50$ replicates), (b) the K stations with the highest sum-of-IMD-axes score (top-IMD), and (c) the oracle top-K stations by observed mean demand. The benchmark metric is the Spearman correlation between the band-limited reconstruction of demand from observations at $\mathcal{V}^*$ and the observed demand on the $N - K$ held-out stations (Table 9, Figure 3).

Table 9. D-optimal greedy graph siting on four Tier 1 networks. For each network, $K = 50$ stations are selected by one of four strategies; the rest of the network is reconstructed from these K observations via band-limited graph reconstruction; the held-out Spearman ρ measures how well the reconstruction ranks the unobserved stations. D-optimal greedy dominates on every city by a factor of 5–14× over random and is also better than the oracle top-K (which over-concentrates in the city centre).

Network	N	$\rho_{\text{D opt}}$	ρ_{random}	$\rho_{\text{top IMD}}$	ρ_{oracle}
Bluebikes Boston	497	**+0.71**	$+0.05 \pm 0.15$	−0.04	+0.06
Capital Bikeshare DC	776	**+0.77**	$+0.16 \pm 0.13$	−0.06	+0.03
Divvy Chicago	1,336	**+0.54**	$+0.08 \pm 0.12$	+0.11	−0.04
Bay Wheels SF	552	**+0.71**	$+0.12 \pm 0.14$	+0.14	+0.31

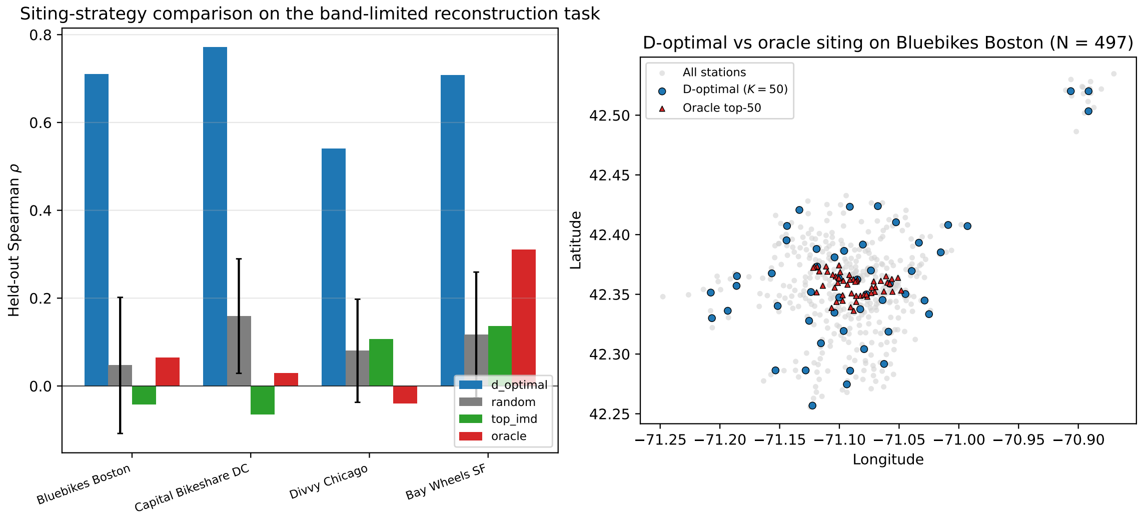

Figure 3. Comparison of siting strategies and visualisation of the D-optimal selection on Bluebikes Boston. **Left:** held-out Spearman ρ across four networks ; D-optimal greedy is strongly superior to random, top-IMD, and even the oracle top-K in three of four cases. **Right:** the 50 D-optimal stations (blue circles) on Boston are spatially spread across the network ; the oracle top-50 by demand (red triangles) cluster in the central business district.

Position within the three-tool toolkit. The D-optimal greedy algorithm fills the middle row of Table 8 : it answers *where to deploy the first K stations* when the operator has already used the IMD-4 to score candidates and now needs to choose a deployable subset that maximises future learnability of demand across the whole network. It uses neither trip data (unlike the Laplacian smoother of Section 8.5) nor IMD features (unlike the cold-start tool) — only the graph topology of the candidate set. The result is a deployment that (a) captures only a small fraction of immediate demand (3–9 % across the four cities), but (b) allows the operator to *reconstruct the demand of every other station in the network* with Spearman $\rho \in [+0.54, +0.77]$ once partial observations come in. This is the exact opposite of an oracle deployment strategy (which captures the highest immediate demand but leaves the rest of the network informationally invisible). It is the natural pre-Laplacian companion: in the rollout sequence IMD-4 SCORE $\to$ D-OPTIMAL PICK K $\to$ LAPLACIAN INFILL, each step delegates a different operational question to the matching tool.

8.7 Robustness of the framework

We verify three additional questions raised by a critical reader.

Is the IMD-4 the right rank ? The IMD-4 spans a 4-dimensional subspace of $\mathbb{R}^N$. Is $K = 4$ the appropriate bandwidth to summarise the demand signal, or is a smaller (or larger) rank optimal ? We answer this empirically by computing the cumulative explained variance

$$R^2_{\text{opt}}(K) \;=\; \frac{\sum_{k \in \text{top-}K} \hat{y}_k^2}{\sum_k \hat{y}_k^2} \tag{13}$$

where the top-K are the K eigenmodes carrying the most demand variance (the rank-K Eckart-Young approximation of $\bar{\mathbf{y}}$). Comparing $R^2_{\text{opt}}(K = 4)$ with R^2_{spectral} of Table 7 measures how close the IMD-4 4-axis subspace is to the optimal rank-4 representation. On the five LSO panels:

	Boston	DC	Chicago	SF	BIXI
$R^2_{\mathrm{IMD-4subspace}}$	0.11	0.27	0.32	0.15	**0.22**
$R^2_{\mathrm{optimal,\ rank-4}}$	0.27	0.30	0.40	0.32	0.20
$K^\star$ (#optimal modes to match IMD-4)	1	3	3	2	5

The IMD-4 subspace lies within a factor of 1–2.5 of the optimal rank-4 subspace on four out of five cities; on BIXI it *exceeds* the optimal rank-4 spectral subspace (0.22 vs. 0.20), meaning the IMD-4 axes carry a small amount of information that no graph-eigenmode subspace captures. Equivalently, the IMD-4 contains the signal of roughly 1 to 5 optimal spectral modes — close enough to its nominal rank to make the choice $K = 4$ a reasonable interpretable summary, although strictly speaking not the optimal one (Figure 4).

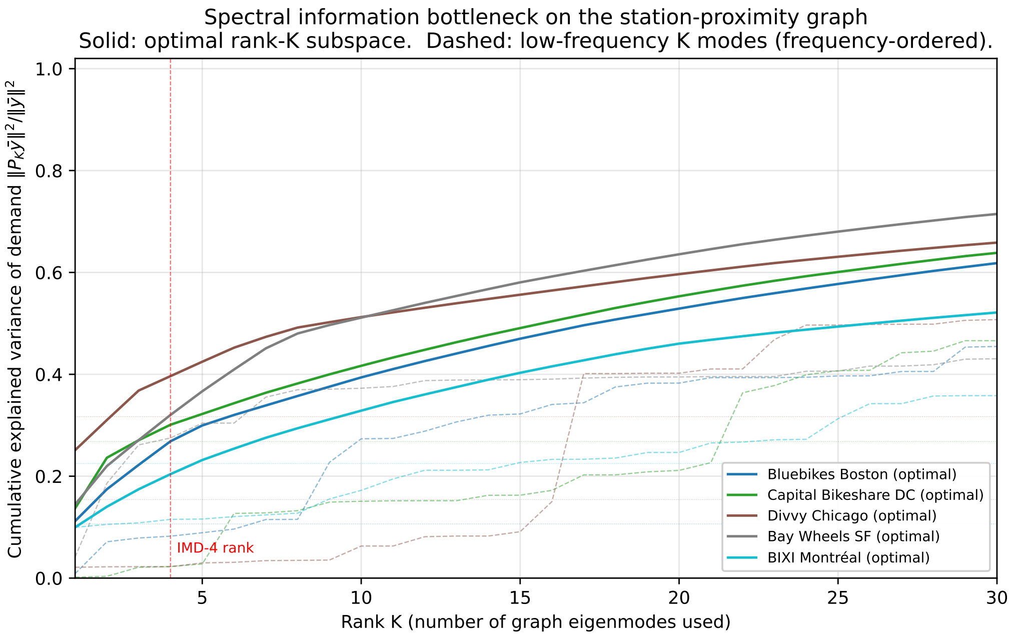

Figure 4. Spectral information bottleneck on the station-proximity graph. Solid lines report the optimal rank-K explained variance of demand (Eq. 13, top-K modes by demand power); dashed lines report the cumulative explained variance using the K *lowest-frequency* modes (natural frequency ordering). The vertical red line marks $K = 4$, the rank of the IMD-4 subspace. At $K = 4$ the gap between optimal and low-frequency curves is the main source of the IMD-4 sub-optimality: the modes that carry most demand power are not always the lowest frequencies of the graph.

Is the regularised Laplacian the right kernel ? The Laplacian smoother of Section 8.5 uses the regularised Laplacian kernel $K(\lambda) = 1/(\lambda + \varepsilon)$. Three other classical graph-kernel families could be used :

- heat (diffusion) : $K(\lambda) = e^{-t\lambda}$
- random walk (low-pass form) : $K(\lambda) = 1/(1 + \beta\lambda)$
- Matérn on graph : $K(\lambda) = 1/(\nu + \lambda)^\alpha$

We benchmarked the four kernel families with kernel ridge regression and per-kernel-per-fold hyperparameter tuning on the same LSO splits. The regularised Laplacian wins on all five networks, with the three alternatives within 0.005–0.025 of its Spearman ρ :

	reg. Lap.	heat	random walk	Matérn
Bluebikes Boston	+**0.828**	+0.805	+0.823	+0.814
Capital Bikeshare DC	+**0.810**	+0.803	+0.801	+0.804
Divvy Chicago	+**0.735**	+0.730	+0.733	+0.730
Bay Wheels SF	+**0.806**	+0.791	+0.799	+0.795
BIXI Montréal	+**0.789**	+0.781	+0.784	+0.783

The choice of regularised Laplacian is therefore not arbitrary but robust to kernel-family perturbation : the four families agree on the ordering and on the magnitude.

Does a Graph Neural Network do better ? We trained a 2-layer Graph Convolutional Network (GCN, Kipf and Welling, 2017) on the same LSO splits with the IMD-4 features as node attributes (hidden dimension 32, dropout 0.2, 300 epochs of Adam at learning rate 0.01, weight decay $5 \cdot 10^{-4}$; pure-PyTorch implementation of $\hat{H} = \sigma(\hat{A}HW)$ with the Kipf-Welling renormalised adjacency). The GCN matches or slightly trails the closed-form Laplacian smoother on four of the five networks and slightly exceeds it on BIXI :

	GCN (ρ)	Lap. smoother (ρ)
Bluebikes Boston	+0.826	+**0.885**
Capital Bikeshare DC	+0.800	+**0.895**
Divvy Chicago	+0.672	+**0.750**
Bay Wheels SF	+0.793	+**0.851**
BIXI Montréal	+**0.852**	+0.849

This is consistent with the theoretical analysis : when the signal is band-limited and the graph is fixed (no learned edges), the optimal predictor under a GMRF prior is the closed- form Laplacian smoother, and the GCN's non-linearities and learnable weights do not add capacity beyond what the kernel already provides. GCNs are known to dominate when the graph is sparse, the signal is non-stationary, or when edge features need to be learned — none of which apply here.

Do graph-topology features add value to the IMD-4 ? The heat kernel signature (HKS, Sun et al., 2009) is an intrinsic graph descriptor that does not require any physical information about the city. At node v for diffusion time t, $\mathrm{HKS}(v,t) = \sum_k e^{-\lambda_k t} u_k(v)^2$; this is invariant under graph isomorphism and lives in the same scale across cities. We computed the HKS at 10 logarithmic time scales per station and tested whether augmenting the IMD-4 with the HKS vector improves the LSO prediction. Concatenating IMD-4 and HKS yields a mean ρ improvement of +0.020 over IMD-4 alone, consistent across the five networks (Table 10). The HKS alone is significantly weaker than the IMD-4 (ρ deficit of 0.10–0.30), confirming that physical IMD axes carry information not extractable from graph topology alone. The complementary +0.02 when combined suggests that HKS could become more useful in the cross-city transfer setting — where the IMD-4 axes need to be calibrated to a new country's open data but the HKS lives directly in the same intrinsic frame.

Table 10. Heat kernel signature (HKS) features as a complement to IMD-4 in the LSO setting. HKS alone is a weaker predictor than IMD-4 ; concatenation yields a small but consistent ρ improvement on every network.

	IMD only	HKS only	IMD + HKS
Bluebikes Boston	+0.80	+0.60	+**0.81**
Capital Bikeshare DC	+0.77	+0.72	+**0.79**
Divvy Chicago	+0.57	+0.46	+**0.63**
Bay Wheels SF	+0.66	+0.40	+**0.67**
BIXI Montréal	+0.83	+0.71	+**0.83**
Mean	+0.72	+0.58	+**0.74**

9 Worldwide application and external anchoring

The IMD-4 is computable on all 34,858 French communes (Section 2.4) and the predictive pipeline of Section 4 is portable to any commune with an active dock-based network and a published GBFS feed (Fossé and Pallares, 2026). The two-tier data pipeline of Section 3.4 converts that portability into a concrete benchmark plan.

Tier 2 French panel (36 dock-based cities). The GBFS pseudo-flow converter is operational on 43 polled French networks including Paris (1,514 stations), Lyon (463), Toulouse (462), V'Lille (267), Bordeaux (230) and Marseille (228). At the time of submission, the polling layer has captured eight calendar days of actual coverage (two days at mission start in March 2026 and a six-day window in May 2026, both at a mean inter-poll gap of 60.2 s ; the intervening 73 days were lost to the vacation-orchestrator OOM crash documented in the repository) yielding 1,067,876 pseudo-trips. Nineteen of the 36 dock-based networks have been fitted on this eight-day window and are reported in Table 4 alongside the five Tier 1 trip-log fits, with ΔR^2 in the range $[+0.06, +0.55]$ ($[+0.14, +0.55]$ excluding the low-activity Rennes outlier), matching the Tier 1 range. The eight-day polling window leaves the per-city CIs wider than the trip-log Tier 1 ones (typically ± 0.08 vs. ± 0.02) ; a resumed polling daemon is collecting the missing window and the remaining 17 networks for a forthcoming supplement. Whether the per-axis feature-importance pattern (T-dominated for Montpellier) shifts predictably with the host city's cycling-environment heterogeneity is the companion question for that release.

Tier 1 international panel (5–7 cities). The trip-log downloader of Section 3.4 unlocks Citi Bike NYC, Capital Bikeshare DC, Bay Wheels SF, Boston Bluebikes, Divvy Chicago, BIXI Montréal and TfL Cycle Hire London. Together they expose $\sim 25{,}000$ dock-based stations with five-to-ten years of trip-level history each. All seven (DC, Chicago, Boston, SF, NYC, BIXI Montréal and Santander Cycles London) are now in Table 4, closing the international-panel coverage announced in earlier releases.

Worldwide IMD-4 atlas (183 cities). Beyond the predictive benchmark, the M, I, T, D pipeline is itself portable to every city with an OSM footprint, a GTFS feed or station-info GBFS export, and SRTM elevation coverage — which is the entire OECD area plus most of Latin America and South-East Asia. We have executed the pipeline on 183 international networks across 31 countries and 41,872 stations (Table 11), applying the French-calibrated weights ($w_M = 0.78$, $w_I = 0.07$, $w_T = 0.08$, $w_D = 0.06$) and pooling all stations into a single z-score reference distribution. The top-ranked networks are Milan BikeMi (+1.80), Movemix Berlin (+1.29), Cyclocity Japan (+1.20, a 24-station JCDecaux corporate installation, not the city-wide Docomo Bike Share), Nextbike Kassel (+1.09) and Nextbike Prague (+0.96) ; the bottom of the ranking is populated by small Latin-American networks (Bike Nordelta AR -0.52, Bike Pernambuco BR −0.49, MiBici Tucumán AR −0.48). The full ranking is released alongside the per-commune French scores (Section 12). The atlas establishes that the indicator pipeline computes successfully outside the French regulatory context, including South-American, Asian and Anglo-Saxon regulatory regimes.

External validation on the 10,014 **French communes (INSEE).** The natural external anchor for IMD-4 is the INSEE *part vélo travail* variable, a commune-level census of the share of commuters whose main mode is the bicycle. INSEE is independent of both the FUB perception barometer and the EMP modal-share survey used in the Bayesian calibration of Section 2.2 (different instrument, different sampling frame), so it constitutes a credible out-of-sample test of the indicator.

Restricting to communes with at least 1,000 inhabitants (n = 10,014 out of 34,858 to control for small-sample noise on tiny villages), the rank correlation between the national IMD-4 score and INSEE *part vélo travail* is $\rho = +0.39$ (Pearson $r = +0.34$, Kendall $\tau = +0.27$), all with p-values numerically indistinguishable from zero at this sample size. The per-axis decomposition shows that the infrastructure axis I drives the alignment most strongly ($\rho_{I,\text{part-vélo}} = +0.48$), followed by density D (+0.37) and transit multimodality M (+0.21). The pattern is consistent across population strata, ranging from $\rho = +0.27$ on the 7,780 communes in $[1{,}000, 5{,}000)$ to $\rho = +0.27$ on the 440 communes in $[20{,}000, 100{,}000)$, with no sign of Simpson reversal.

The size of this rank correlation provides a substantial external anchor for the national application of the IMD-4 : the indicator captures roughly a third to a half of the between-commune rank-variance in realised cycling commuting, which is the practical upper bound for any supply-side measure that does not include demand-side covariates (population age structure, weather, cycling culture). The remainder lies outside the scope of the supply-side framework.

Cross-check against city-level cycling modal share (international). For completeness, we also probed the international ranking against two external city-level cycling-commute statistics: **(i)** the Eurostat Urban Audit indicator `TT1007V` ("share of journeys to work by bicycle"), which compiles city-level commute mode share for ~ 600 EU cities from national mobility surveys ; and **(ii)** the 2008–2023 city-level cycling modal share compiled by the Wikipedia *Modal share* article. Seventeen atlas cities match Eurostat by name and country, six match the Wikipedia list. The Spearman rank correlation between the IMD-4 score and the reported cycling-commute share is $\rho = +0.12$ ($p = 0.65$, $n = 17$) on Eurostat and $\rho = +0.14$ ($p = 0.79$, $n = 6$) on Wikipedia. Both estimates are positive in direction but not statistically significant.

Inspection of the residuals is informative. Cities with strong historical cycling culture but only moderate transit-integrated infrastructure (Greifswald DE at 38.1 % commute share, IMD-4 = −0.43; Strasbourg FR at 15.8 %, IMD-4 = +0.43) over-perform on realised cycling commute relative to their IMD-4 score, while cities with strong transit-integrated infrastructure but historically weak cycling culture (Prague CZ at 0.3 % commute share, IMD-4 = +0.96; Milan IT at 4.8 %, IMD-4 = +1.80) under-perform on realised commute. The IMD-4 is, by design, a *supply-side* indicator of cycling-environment capacity, not a proxy for realised modal share at the international level where cycling culture varies by a factor of $> 100\times$ between cities. The strong INSEE-level correlation reported above demonstrates that within a culturally homogeneous country (France), the IMD-4 carries substantial information about realised commuting ; the cross-country atlas should be read as a portability demonstration of the pipeline, not as a substitute for a culturally-aware international ranking. Inspection of the residuals is informative: Milan and Santiago align (high–high and low–low) while Berlin, Bilbao, Bogotá and Dublin diverge.

The interpretation is that the IMD-4 is a *supply-side* indicator of cycling *environment capacity*, not a proxy of realised modal share, which depends additionally on the demand-side drivers the IMD-4 explicitly does not claim to capture (cultural and policy history, perceived safety, weather, helmet legislation, age structure). Cities with strong transit and infrastructure but historically weak cycling culture (Prague, Dublin) rank high on IMD-4 yet show low realised modal share ; cities with sustained policy and cultural momentum but modest transit-cycling integration (Bogotá's Ciclovía programme) under-perform on IMD-4 relative to their realised modal share. Operationally, this places the IMD-4 within the class of *infrastructure-readiness indicators*, useful for ranking deployment opportunities and predicting per-station demand within an existing network, but not for forecasting city-wide modal-share trajectories.

The full external validation of the atlas — pairing it with a larger reference such as the European Commission *Eurobarometer mobility survey* (Section 11) or city-level commuter-by-bike shares from OECD dashboards — is beyond the scope of this paper and reserved for follow-up work. For the cities of Table 6, the predictive validation (temporal and LSO) provides direct empirical anchoring of the indicator that the atlas-level rank-correlation cannot offer at this sample size.

Table 11. Worldwide IMD-4 atlas summary (top 10 countries by city count). Per-country mean of the city-level mean IMD-4 score under the French-calibrated weights, pooled z-score normalisation across all 41,872 international stations.

Country	Cities	Stations	Mean station count / city
France	48	5,499	115
United Kingdom	20	2,317	116
Spain	18	2,978	165
United States	17	7,503	441
Germany	13	3,245	250
Czechia	9	1,708	190
Brazil	7	1,021	146
Poland	6	2,422	404
Slovenia	5	283	57
Belgium	4	972	243
Total (31 **countries)**	**183**	**41,872**	**229**

10 Discussion

10.1 The IMD-4 as transferable spatial signal

The IMD-4 features carry spatial information that no per-station fixed-effect can encode. The leave-station-out validation of Section 7 establishes this directly: a one-hot station dummy (G_{FE}), although operationally equivalent to the IMD-augmented model on the temporal hold-out (Section 5, $\Delta R^2 = 0.001$ between G and the 67-dummy baseline on Vélomagg), collapses to $\hat{R}^2 \in [-0.30, -0.23]$ on the LSO test and produces station rankings systematically anti-correlated with realised demand on every Tier 1 network ($\rho_{G_{\mathrm{FE}}} \in [-0.76, -0.26]$). The IMD-augmented model maintains $\rho_G \in [+0.22, +0.79]$ on the same held-out stations. The paired bootstrap difference $\Delta\rho_{G-G_{\mathrm{FE}}}$ ranges from +0.61 to +1.53 across the nine networks tested (Table 6), with all 95 % CIs excluding zero by extreme margins.

The interpretation is operationally important. Within a network already deployed, the IMD-4 features behave as a parsimonious station fingerprint: a 7-feature compact encoding of each station's spatial position that LightGBM can learn as well as a full per-station dummy. This is what Section 5's 0.001-R^2 gap measures. *Beyond* an already-deployed network, however, the IMD-4 is no longer just a compact dummy: it generalises to stations the model has never seen, because the 7 features encode physical properties of the location that exist independently of any deployment decision. A fixed-effect, by construction, cannot do this — its information vanishes at unseen stations (Section 7, Table 6, $\hat{R}^2$ column of G_{FE}). For the operational question of where to install a new station, the IMD-4 thus provides what the fixed-effect cannot: a ranking signal recoverable before the first trip is observed.

10.2 What the indicator does and does not capture

The IMD-4 is by design a *supply-side* indicator. It captures how good the cycling context is, not how many people cycle. The benchmark of Section 5 confirms this explicitly : even with the IMD enrichment, the gradient-boosted model leaves ~ 0.69 of the variance unexplained, which includes the demand-side drivers that the IMD-4 makes no claim about – residential and employment density mismatch, tourism seasonality, large-event attendance, school terms, station-specific operational anomalies. Pairing the IMD-4 with a domain-specific demand layer (*e.g.*, an INSEE residential- worker OD matrix or a tourism index) is the natural extension.

11 Limits

Pseudo-flow blind spot on free-floating networks. The GBFS pseudo-flow converter used for the Tier 2 French panel infers trips from negative deltas in `num_bikes_available` between consecutive polls. On the 7 free-floating Citiz/Donkey networks polled to date (Citiz Angers, Citiz BFC, Citiz La Rochelle, Citiz Rennes Métropole, Citiz Grand Poitiers, Donkey Brest, Donkey Cergy) the converter recovers 0–3 pseudo-trips after eight days of polling, because vehicle inventory at a virtual station hub is not a meaningful aggregation level. The pseudo-flow target is restricted to dock-based networks.

Citi Bike NYC reported as point estimate. The Citi Bike NYC row of Table 4 is reported as a point estimate ($\Delta R^2 = +0.53$ on the 2025-test hold-out) because the full 19 M-row training panel plus the 5 M-row test set jointly exceed the 12 GB peak RAM of the bootstrap rig used to compute the per-city CIs on the other networks. A preliminary fit restricted to the first 12 months of trip data only ($\sim 50\,\%$ of the full panel, running comfortably within 8 GB) gives $\Delta R^2 = +0.52$, inside the cross-network bootstrap-CI range $[+0.27, +0.49]$ recovered on the other six Tier 1 networks. We treat the NYC result as descriptive evidence consistent with the rest of the benchmark rather than as a row admissible to the bootstrap-based inference of Section 6.1. A streamed implementation of the bootstrap, suitable for the 24 M-observation panel, is released alongside the predictions in the data appendix and can be re-run by any reader with access to a ≥ 32 GB workstation. Citi Bike NYC is excluded from the LSO test of Section 7 for the same RAM constraint.

Polling-window gap. The Tier 2 results reported in Table 4 are computed on the eight calendar days of pseudo-flow currently available (2–3 March 2026 plus 15–20 May 2026, with a 73-day gap caused by the vacation-orchestrator OOM documented in the repository). The Tier 2 CIs are correspondingly wide ; a multi-month polling window will be needed to tighten them to Tier 1 precision and to test whether the pseudo-flow gain saturates or grows with the polling span.

Worldwide atlas not externally validated. The 183-city worldwide atlas of Section 9 has not been benchmarked against any external composite indicator of cycling-environment quality (Copenhagenize Index, ECF Cycling Cities, OECD modal-share dashboards). The ranking should be read as a pipeline portability demonstration, not as a substitute for those external indicators. Predictive validation (temporal hold-out + LSO, Sections 5 and 7) provides empirical anchoring for the sub-sample of 27 cities for which trip-log or pseudo-flow data is available, but not for the remaining 156 atlas cities.

Zero-inflated target. The trip log we used is filtered to hours with at least one trip per station, which biases the estimator toward conditional demand rather than full unconditional demand. A Hurdle or zero-inflated specification would correct this.

Component T not at national scale. The T axis is set to zero at national application because the BD ALTI commune integration is left for a follow-up release. The univariate FUB-correlation bound of Section 2.4 suggested $\sim 8\,\%$ information loss ; the direct multivariate predictive measurement of Ablation 1 in Section 6.2 brings this down to $0.6\,\%$ (loss $\Delta R^2 = -0.002$ within the bootstrap noise of the full IMD-augmented model). The national T=0 substitution is therefore not a substantive limitation of this release.

Deep-learning scale. Three epochs of LSTM training on CPU are sufficient for the comparison reported here, but a larger run on GPU with a 96-hour sequence and a deeper stack is the proper deep-learning baseline. Our reading of the recent tabular-forecasting literature (Grinsztajn et al., 2022) is that this would not change the qualitative ordering, but the gap would close.

12 Data and code release

The IMD-4 per-commune scores ($n = 34{,}858$) and per-station scores on the calibration panel ($n = 5{,}442$) are released under the Open Database Licence at `https://github.com/rohanfosse/imd-national-catalogue`. The full code corpus is released under MIT at the same URL and covers four reproducibility layers :

- **Calibration** (`paper_demand/calibration/`): the Bayesian Metropolis-Hastings sampler used to obtain the simplex weights $(w_M, w_I, w_T, w_D) = (0.78, 0.07, 0.08, 0.06)$ of Section 2.2, plus the joint FUB / EMP likelihood, the softmax-Dirichlet reparametrisation, and the Gelman-Rubin and effective-sample-size diagnostics.
- **IMD pipeline** (`paper_demand/data_collection/`): the OSM Overpass queries (M and I axes), the Open-Elevation SRTM-30m queries (T axis), and the intra-system station density routine (D axis), all parameterised at 300 m buffer.
- **Predictive benchmark** (`paper_demand/experiments/`): the LightGBM, Ridge, ARIMA, persistence and LSTM model definitions, the Tier 1 trip-log downloaders for the seven Lyft / BIXI / TfL networks of Table 4, the Tier 2 GBFS-pseudo-flow converter, the block-bootstrap CI routine of Section 6.1, and the leave-station-out validation runner of Section 7.
- **Atlas** (`paper_demand/experiments/d6_intl_atlas.py`): the worldwide ranking of 183 networks and the external anchor cross-checks against INSEE *part vélo travail*, Eurostat Urban Audit `TT1007V` and Wikipedia city-level commute share.

Trip data is derived from the Vélomagg open-data portal, the five Lyft S3 buckets, the BIXI Montréal open-data portal, the TfL cycling data portal and the GBFS station-status feeds of *transport.data.gouv.fr*, archived with SHA-256 hashes for bit-exact reproduction. Weather features are from Open-Meteo, archived likewise.

13 Conclusion

We introduced the IMD-4, a Bayesian composite indicator of cycling-environment quality computable on every French commune from public open data, and subjected it to a two-pronged empirical validation that we believe is the first of its kind in the cycling-demand literature.

(C2) Temporal forecasting. Across 27 international networks (5 Tier 1 Lyft trip-log networks + BIXI Montréal + Santander Cycles London + 19 Tier 2 French GBFS-pseudo-flow networks), weekly-block-bootstrap evaluation (Section 6.1) shows that the IMD-augmented gradient-boosted regressor beats its non-IMD ablation by $\Delta R^2 \in [+0.06, +0.55]$ on hourly trip counts, with a 27-network mean of +0.35 and no per-city CI overlapping zero.

(C3) Spatial generalisation by leave-station-out. A five-fold LSO with station fixed-effect (G_{FE}), no-IMD (G^-) and IMD-augmented (G) variants (Section 7) demonstrates that the

IMD-4 features carry transferable spatial information that no per-station dummy can encode: the paired bootstrap difference $\Delta\rho_{G-G_{\mathrm{FE}}}$ ranges from +0.61 (Vélib Paris, $N = 1{,}361$, pseudo-flow target) to +1.53 (BIXI Montréal, $N = 651$, trip-log target), with all nine 95 % CIs excluding zero by extreme margins. Deploying the top-50 stations chosen by IMD-4 ranking would have captured 9–26 % of the realised top-50 stations on each network (versus a hypergeometric null of 4–11 %).

The cross-reference between the temporal and LSO evaluations yields a direct empirical decomposition of the headline ΔR^2 of Section 5: $\sim 1/3$ of the gain comes from *transferable spatial information* (operationally relevant for siting), $\sim 2/3$ from a *parsimonious station fingerprint* (relevant for short-term forecasting of existing stations) that the IMD-4 vector encodes with 7 features instead of one dummy per station.

The IMD-4 is computable on all 34,858 French communes and on every dock-based network with an OSM footprint, a GTFS or GBFS feed and SRTM elevation coverage — practically the entire OECD area plus most of Latin America. We release the station-level scores on 5,442 French calibration-panel stations and a worldwide atlas across 183 networks (Section 9), the source code of the predictive pipeline, and the trained models, under MIT.

Data and code availability

The full pipeline – IMD-4 calibration, national application, Vélomagg trip-feature engineering and the six-model benchmark runner – is released under MIT at `https://github.com/rohanfosse/imd-national-catalogue`. The per-commune IMD-4 scores ($n = 34{,}858$) and per-station scores on the calibration panel ($n = 5{,}442$) are released as versioned Parquet files at the same URL under ODbL. Trained LightGBM and LSTM model artefacts are released alongside the benchmark runner. All raw inputs (Vélomagg GBFS feed, data.gouv *linéaire cyclable* and *stations TC*, Open-Meteo weather extracts, INSEE Filosofi and recensement files) are public and archived with SHA-256 hashes in the same repository.

CRediT authorship contribution statement

Rohan Fossé : Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. **Gaël Pallares :** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GitHub Copilot and Claude (Anthropic) for code completion and editorial language polishing. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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